华中科技大学启明选拔考试合集

学数华科　转专业交流群

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写在一切之前

首先，如果你计划通过启明选拔进入实验班，建议完整阅读《华科转专业简明指南（2025.08.27）》，并以学校当年发布的招生政策和通知为准。本章内容参考该版本指南整理，原文版权归相关作者所有。

如果你是在华科转专业交流群之外获取到本合集，也欢迎加入指南所列的华科转专业交流群（QQ 群：369787431），或加入 2025 级学数华科（QQ 群：920066268），查阅政策汇总、往年真题和经验分享。

关于启明选拔

开学期间举行的实验班选拔考试通常被称为“启明选拔”。通过这一途径，可以申请进入其他学院或专业的实验班，也可以申请进入本学院的其他实验班。各年度开放选拔的专业、报名条件和考查方式并不完全相同，多数专业还设有高考选科限制，应以当年招生计划和学校通知为准。

启明选拔通常分为统考笔试和学院面试。笔试一般安排在新生军训期间，主要考查数学、综合和英语；各学院根据初试成绩确定复试名单，复试多采用面试形式。所有形式的转专业原则上只能成功一次；通过启明选拔进入实验班后，通常不能再报名当年 12 月的转专业统一考试。因此，报名前应充分了解目标专业的培养方案、课程压力和后续发展。

关于本合集

本合集按照“数学、综合、英语”的顺序分为三个部分，并在各部分内按年份由近到远编排。不同年份的资料完整程度不一，部分内容由往届同学依据考后回忆和现场记录整理，可能与正式试卷存在差异。

- **数学：**考试范围与高中数学竞赛相近。复习时可先熟悉往年真题，并保持计算、推理和综合运用能力。
- **综合：**题型和内容可能逐年变化，通常结合现实背景考查阅读理解、信息提取、数学建模、逻辑分析和临场应变能力。
- **英语：**难度与大学英语六级相近，题型与素材可能随年份调整。复习时应重视词汇辨析、篇章理解、上下文逻辑和写作表达。

真题和答案仅供复习参考，不宜将回忆版中的表述或参考答案视为官方结论。若发现错漏，或持有更清晰的原卷、缺失题目和可靠答案，欢迎通过邮箱@那，边。向我反馈，以便后续勘误和更新。

关于实验班

实验班并不等同于“重点班”，也未必适合每一位同学。部分实验班设有较明确的推免保障、导师指导或科研训练安排，但具体成绩、课程和无挂科要求由各学院制定；与此同时，实验班的课程进度和难度往往更高，也可能影响加权成绩、外校推免或出国申请。选择前应尽量联系往届同学，了解课程安排、培养方向和真实就读体验。

另外，本文件已上传至《华中科技大学启明选拔考试合集》，大家使用前可以先检查一下是否有更新。

最后，祝大家选拔顺利、得偿所愿！

CONTENTS

Part I

启明考试数学部分

CHAPTER 1

2025 年启明考试数学部分

致谢： 本套试题来自 @ 鳄鱼，感谢 @ 鳄鱼同学提供的完整试题！

1　填空题 (每小题 12 分，共 72 分)

1. 设整数数列 $a_1, a_2, \ldots, a_{10}$ 满足 $a_{10} = 3a_1$，$a_2 + a_8 = 2a_5$，且 $a_{i+1} \in \{1 + a_i, 2 + a_i\}$，$i = 1, 2, \ldots, 9$. 这样数列 n 的个数为[1]__________.
2. 若非空整数集 $S \subseteq \{1, 2, 3, 4, 5\}$，那么符合条件“若 $x \in S$，则 $y - x \in S$”的集合 S 的个数是[2]__________.
3. 若 α, β, γ 为正数，且 $3\alpha^2 + \beta^2 + \gamma^2 = 1$，则 $\dfrac{\beta + 1}{\alpha\beta\gamma}$ 的最小值__________.
4. 若 u, v, w 为复数，$|u| = |v| = |w| = 1$ 且 $u + v + w = 1 + i$，则 u 实部最小值为__________.
5. 设 α, β 满足
$$x \sin\beta + y \cos\alpha = \sin\alpha, \quad x \sin\alpha + y \cos\beta = \sin\beta,$$
则 $x^2 - y^2$ 的值为[3]__________.
6. 正方形 $ABCD$，点 P 满足 $PA : PB : PC = 1 : 2 : k$，则 k 的可能取值为__________.

2　解答题 (本题 12 分)

求方程 $x^3 - 3x + 1 = 0$ 的解.

3　解答题 (本题 15 分)

设数列 $\{a_n\}$ 满足 $a_1 = 3$，$a_{n+1} \le \dfrac{1}{2}(a_n + a_{n+2})$，$a_{505} = 2145$，求 a_5 的最大值.[4]

[1] 本题与 2019 年启明数学填空题第 **5** 题一致.

[2] 原题中的 y 未作定义，因而条件不足，无法唯一确定答案.

[3] 本题与 2019 年启明数学填空题第 **9** 题一致.

[4] 本题为 2019 年启明数学第 **3** 题的变式.

CHAPTER 2

2025 年启明考试数学部分参考答案

1　填空题 (每小题 12 分，共 72 分)

1. 参见 2019 年启明数学填空题第 5 题**解析**.

2. **Solution**. 原题中的 y 未作定义，因此无法唯一作答.
以下列出三种常见的可能修订，供核对原题时参考.
一种可能是原题中的 y 为数字 6 的笔误；此外也可能缺少对 x, y 的限定.

情形一. 若条件为 “$x \in S \Rightarrow 6 - x \in S$”,
条件 “若 $x \in S$, 则 $6 - x \in S$” 意味着元素必须成对出现. 我们将集合 $\{1, 2, 3, 4, 5\}$ 分成满足此条件的块:
1. 若 $1 \in S$, 则 $6 - 1 = 5 \in S$. 反之亦然. 所以 $\{1, 5\}$ 是一个块.
2. 若 $2 \in S$, 则 $6 - 2 = 4 \in S$. 反之亦然. 所以 $\{2, 4\}$ 是一个块.
3. 若 $3 \in S$, 则 $6 - 3 = 3 \in S$. 这自然满足. 所以 $\{3\}$ 是一个块.
任何满足条件的集合 S 都是这些块的并集，其中

$$B_1 = \{1, 5\}, \qquad B_2 = \{2, 4\}, \qquad B_3 = \{3\}.$$

因此，只需考察由 $\{B_1, B_2, B_3\}$ 组成的非空集合.
可能的集合是 $\{B_1, B_2, B_3\}$ 的任何非空子集的并集. 子集的数量为 $2^3 - 1 = 7$.
它们是: $\{3\}, \{1, 5\}, \{2, 4\}, \{1, 3, 5\}, \{2, 3, 4\}, \{1, 2, 4, 5\}, \{1, 2, 3, 4, 5\}$.

情形二. 若条件为 “$x, y \in S$ 且 $x \neq y \Rightarrow y - x \in S$”,
若 $|S| \geq 2$，取 $a < b \in S$。由条件对有序对 $(x, y) = (b, a)$ 也必须成立，故 $a - b \in S$。但 $a - b < 0$，而 $S \subseteq \{1, 2, 3, 4, 5\}$ 不可能含负数，矛盾。因此 $|S| = 1$，答案为 5 种

情形三. 若条件为 “$x, y \in S$ 且 $y > x \Rightarrow y - x \in S$”,
若 $1 \in S$，则由 $a \in S \Rightarrow a - 1 \in S$ 递推，得到 $S = \{1, 2, \ldots, k\}$ 型，且 $k = 1, 2, 3, 4, 5$ 均可。
若 $1 \notin S$，则不能同时含有相邻数（否则差为 1）；进一步可证此时 $|S| \leq 2$。检验两元素情形，只剩 $\{2, 4\}$ 满足（如 $\{2, 3\}$ 会强制 $1 \in S$；$\{3, 5\}$ 会强制 $2 \in S$ 等）。
单元素集合均真命题（无 $y > x$ 的成对元素），因此 $\{1\}, \ldots, \{5\}$ 都成立。合计 10 种

3. **Solution**. $8\sqrt{3}$
这道题最简单方法是拉格朗日乘子法，不过考虑新生没有学就用另外的方法计算。
不妨设 $x = \sqrt{3}\,\alpha$, $y = \beta$, $z = \gamma$，则 $x, y, z > 0$ 且

$$x^2 + y^2 + z^2 = 1, \qquad \frac{\beta + 1}{\alpha\beta\gamma} = \sqrt{3}\,\frac{y + 1}{xyz}.$$

对固定的 y，由 $x^2+z^2=1-y^2$ 及 $xz\le\dfrac{x^2+z^2}{2}$ 得 $xyz\le y\cdot\dfrac{x^2+z^2}{2}=\dfrac{y(1-y^2)}{2}$, 于是有

$$\frac{\beta+1}{\alpha\beta\gamma}=\sqrt{3}\,\frac{y+1}{xyz}\ge\sqrt{3}\,\frac{y+1}{\frac{y(1-y^2)}{2}}=2\sqrt{3}\,\frac{y+1}{y(1-y^2)}.$$

下面只需求 $\min\limits_{0<y<1}\dfrac{y+1}{y(1-y^2)}$, 易知当 $y=\dfrac{1}{2}$ 取最小值 4。故

$$\frac{\beta+1}{\alpha\beta\gamma}\ge 2\sqrt{3}\cdot 4=8\sqrt{3}.$$

取"="：由 $y=\dfrac{1}{2}$ 且 $xz=\dfrac{1-y^2}{2}=\dfrac{3}{8}$，又为使 xz 最大有 $x=z=\sqrt{\dfrac{3}{8}}$，故

$$\beta=\frac{1}{2},\quad \sqrt{3}\,\alpha=x=\sqrt{\frac{3}{8}}\ \Rightarrow\ \alpha=\frac{1}{2\sqrt{2}},\quad \gamma=z=\sqrt{\frac{3}{8}}.$$

代回可取得等号且有最小值 $8\sqrt{3}$

4. **Solution**. $\dfrac{-1-\sqrt{7}}{4}$
不妨设 $u=x+iy$. 我们要最小化 x.
由 $u+v+w=1+i$, 得 $v+w=(1+i)-u=(1-x)+i(1-y)$. 利用三角不等式, $|v+w|\le|v|+|w|=1+1=2$.
所以 $|v+w|^2\le 4$. $|v+w|^2=(1-x)^2+(1-y)^2=1-2x+x^2+1-2y+y^2$. 因为 $|u|=1$, 所以 $x^2+y^2=1$. $|v+w|^2=1-2x+1+1-2y=3-2x-2y$. 我们有 $3-2x-2y\le 4$.
设 $x=\cos\theta, y=\sin\theta$. $3-2\cos\theta-2\sin\theta\le 4\implies -1\le 2(\cos\theta+\sin\theta)=2\sqrt{2}\sin(\theta+\frac{\pi}{4})$. $\sin(\theta+\frac{\pi}{4})\ge-\dfrac{1}{2\sqrt{2}}$.
我们要最小化 $x=\cos\theta$. 设 $\phi=\theta+\frac{\pi}{4}$. $x=\cos(\phi-\frac{\pi}{4})=\cos\phi\cos\frac{\pi}{4}+\sin\phi\sin\frac{\pi}{4}=\dfrac{\sqrt{2}}{2}(\cos\phi+\sin\phi)$.
为使 x 最小, 我们需要 ϕ 在第三或第四象限. 当 $\sin\phi=-\dfrac{1}{2\sqrt{2}}=-\dfrac{\sqrt{2}}{4}$ 时, $\cos\phi=-\sqrt{1-\sin^2\phi}=-\dfrac{\sqrt{14}}{4}$.
此时 x 取得最小值: $x=\dfrac{-1-\sqrt{7}}{4}$.
另解：设 $s=1+i$。存在单位复数 v,w 使 $v+w=s-u$ 的充要条件是 $|s-u|\le 2$。需在

$$|u|=1,\quad |s-u|\le 2$$

下最小化 $\Re(u)$。极值在边界 $|s-u|=2$ 与 $|u|=1$ 的交点处。设 $u=(x,y)$，则

$$x^2+y^2=1,\qquad (x-1)^2+(y-1)^2=4\implies x+y=-\tfrac{1}{2}.$$

联立得 $2x^2+x-\frac{3}{4}=0$，故

$$x=\frac{-1\pm\sqrt{7}}{4},\quad \min\Re(u)=-\frac{1+\sqrt{7}}{4}.$$

5. 参见 2019 年启明数学填空题第 9 题**解析**.

6. **Solution**. $k\in\left[2\sqrt{2}-1,\ 2\sqrt{2}+1\right]$
取正方形边长为 1，设 $A(0,0)$, $B(1,0)$, $C(1,1)$，点 $P(x,y)$，记 $r=PA$。由

$$PA^2=x^2+y^2=r^2,\qquad PB^2=(x-1)^2+y^2=4r^2$$

得

$$(x-1)^2+y^2=4(x^2+y^2) \;\Rightarrow\; 3x^2+3y^2+2x-1=0 \;\Leftrightarrow\; (x+\tfrac{1}{3})^2+y^2=\left(\frac{2}{3}\right)^2,$$

又

$$PC^2=(x-1)^2+(y-1)^2=x^2+y^2-2x-2y+2,$$

从而

$$k^2=\frac{PC^2}{PA^2}=1+\frac{2(1-x-y)}{x^2+y^2}.$$

在圆上作参数化 $x=-\frac{1}{3}+\frac{2}{3}\cos t,\ y=\frac{2}{3}\sin t$ 可得

$$k^2=1+\frac{2\left(1--\frac{1}{3}+\frac{2}{3}(\cos t+\sin t)\right)}{\frac{5}{9}-\frac{4}{9}\cos t}=\frac{12\sin t+16\cos t-29}{4\cos t-5}.$$

对 t 求导并令零，或用三角恒等式配角化简，可知极值于两处取得，分别给出 $k_{\min}=2\sqrt{2}-1$，$k_{\max}=2\sqrt{2}+1$； 故 $k\in[2\sqrt{2}-1,\ 2\sqrt{2}+1]$，端点可达。

2 解答题 (本题 13 分)

Solution. $x_1=2\cos\left(\frac{2\pi}{9}\right), x_2=2\cos\left(\frac{4\pi}{9}\right), x_3=2\cos\left(\frac{8\pi}{9}\right)$

对于方程 $x^3-3x+1=0$, 这是一个典型的不可约三次方程, 可以使用三角函数法求解. 不妨设 $x=2\cos\theta$. 将其代入方程中:

$$(2\cos\theta)^3-3(2\cos\theta)+1=0$$

$$8\cos^3\theta-6\cos\theta+1=0$$

$$2(4\cos^3\theta-3\cos\theta)+1=0$$

使用三倍角公式 $\cos(3\theta)=4\cos^3\theta-3\cos\theta$, 方程变为:$\cos(3\theta)=-\frac{1}{2}$

这个方程的解是 $3\theta=\pm\frac{2\pi}{3}+2k\pi$, 其中 k 是任意整数. 因此, $\theta=\pm\frac{2\pi}{9}+\frac{2k\pi}{3}$. 为了得到三个不同的 x 值, 我们可以取 $k=0,1,2$:

所以, 方程的三个实数解是:

$$x_1=2\cos\left(\frac{2\pi}{9}\right),\quad x_2=2\cos\left(\frac{4\pi}{9}\right),\quad x_3=2\cos\left(\frac{8\pi}{9}\right)$$

注: 韦达定理，三次方程求根公式亦可求解本题

3 解答题 (本题 15 分)

Solution. 20

分析：一个自然的想法是所有的等号均能取得时 a_5 取得最大值，此时 $\{a_n\}$ 为 $d=\frac{17}{4}$ 的等差数列，解得 $a_5=20$

不妨设 $b_n=a_{n+1}-a_n, a_{n+1}\le\frac{1}{2}(a_n+a_{n+2})\Rightarrow b_n\le b_{n+1}$

$\therefore a_5=a_1+b_1+b_2+b_3+b_4$

$a_{505}-a_1=\sum_{i=1}^{504}b_i\ge 126(b_1+b_2+b_3+b_4)\Rightarrow b_1+b_2+b_3+b_4\le 17$

$\therefore a_5=a_1+b_1+b_2+b_3+b_4\le 20$, 当 $\{a_n\}$ 为等差数列取 ”$=$”

CHAPTER 3

2024 年启明考试数学部分

1 填空题 (每小题 11 分，共 77 分)

1. 若 $\alpha,\beta,\delta,\gamma(\alpha>\beta>\delta>\gamma)$ 为整数，且 $2^{\alpha}+2^{\beta}+2^{\delta}+2^{\gamma}=20\dfrac{5}{8}$，则 $(\alpha+\beta+\delta+\gamma-1)^{2024}$ 的值为__________.
2. $\left(\dfrac{\sqrt{5}+1}{2}\right)^{12}$ 的整数部分为[1]__________.
3. 设 $f(x)=(2x^5+2x^4-53x^3-57x+54)^{2023}$，则 $f\left(\dfrac{\sqrt{111}-1}{2}\right)$ 的值为__________.
4. 集合 $\Delta=\{1,2,3,5,6\}$ 的全部非空子集的元素和为__________.
5. 在 $(-1,1)$ 上任意取两个数，则这两个数之和小于 0.4 的概率为__________.
6. 直三棱柱 $ABC-A'B'C'$，侧棱长 4，底面边长都为 2，过 A 点截面交 BB',CC' 于 M,N，若 ΔAMN 为直角三角形，求三角形面积最大值__________.
7. 若 $\theta\in\left(-\dfrac{\pi}{2},\pi\right)$，则 $\dfrac{2\sin\theta+4\cos\theta+5}{\sin\theta+\cos\theta+1}$ 的最小值为__________.

2 证明题 (本题 10 分)

证明：若对于 $\forall x,y\in(-\infty,+\infty)$，有 $|f(x)-f(y)|\le|x-y|^2$，则对于每个正整数 n，设 $\forall a,b\in(-\infty,+\infty)$，有 $|f(b)-f(a)|\le\dfrac{|b-a|^2}{n}$。

3 证明题 (本题 13 分)

在数学中，广义距离是泛函分析中最基本的概念之一。对平面直角坐标系中的两个点

$$P_1(x_1,y_1),\qquad P_2(x_2,y_2),$$

记

$$|P_1P_2|_t=\max\left\{\frac{|x_1-x_2|}{1+|x_1-x_2|},\frac{|y_1-y_2|}{1+|y_1-y_2|}\right\}$$

称 $|P_1P_2|_t$ 为点 P_1 与点 P_2 之间的“t-距离”，其中 $\max\{p,q\}$ 表示 p,q 中较大者。

[1] 本题与 2022 年启明数学填空题第 1 题一致.

证明：对任意点 $P_1(x_1,y_1)$，$P_2(x_2,y_2)$，$P_3(x_3,y_3)$，

$$|P_1P_3|_t \leq |P_1P_2|_t + |P_2P_3|_t$$

CHAPTER 4

2024 年启明考试数学部分参考答案

1　填空题 (每小题 11 分，共 77 分)

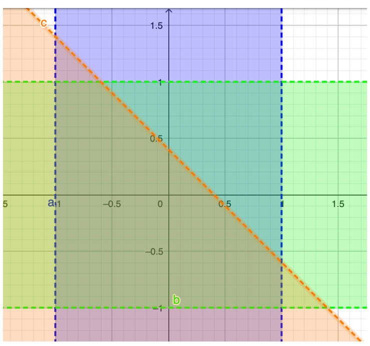
1. **Solution**. 1
注意到 $2^{\alpha}+2^{\beta}+2^{\delta}+2^{\gamma}=\dfrac{165}{8}\Rightarrow 2^{\alpha+3}+2^{\beta+3}+2^{\delta+3}+2^{\gamma+3}=165$ 现在需要找到一组整数 $\alpha,\beta,\delta,\gamma$ 使得方程成立, 注意到：$165=2^7+2^5+2^2+2^0$ 解得：$\alpha=4,\beta=2,\delta=-1,\gamma=-3$ 故 $(\alpha+\beta+\delta+\gamma-1)^{2024}=1$。

2. 参见 2022 年启明数学填空题第 1 题**解析**.

3. **Solution**. -1
令 $t=\dfrac{\sqrt{111}-1}{2}$，则 $2t^2+2t-55=0$。
故 $f(t)=((2t^2+2t-55)(t^3+t-1)-1)^{2023}=(-1)^{2023}=-1$。

4. **Solution**. 272
对于集合中的每个元素, 它在所有子集中出现的频率是相同的。由于集合的大小是 5, 每个元素在一半的非空子集中都会出现。集合的非空子集总共有 $2^5-1=31$ 个，而每个元素将出现在 $2^{(5-1)}=16$ 个子集中。
每个元素在一半的子集中出现，因此总和可以通过将每个元素乘以其出现次数 16 并相加来计算最终结果：$16\times(1+2+3+5+6)=272$

5. **Solution**. 0.68
在（-1,1）上任取 2 个数，设两数为 x,y，作图：



两数之和小于 0.4，$\therefore\begin{cases}-1<x<1\\-1<y<1\\x+y<0.4\end{cases}$

因此两数之和小于 0.4 的概率为

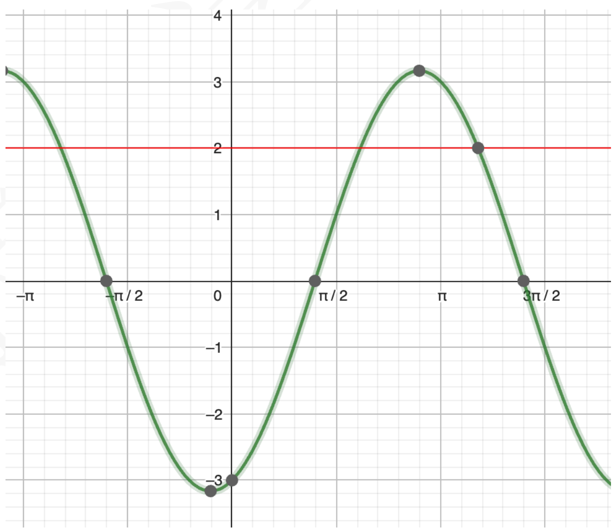
$$p = \frac{4 - \frac{1}{2} \times 1.6^2}{4} = 1 - 0.32 = 0.68.$$

6. **Solution**. $\sqrt{17}$
设 $CM = x$，$BN = y$，$x, y \in [0, 4]$。不妨设 $\angle AMN = 90°$，则 $AN^2 = MN^2 + AQ^2$，即 $4 + y^2 = 4 + x^2 + 4 + (y - x)^2$，整理得：$x^2 - xy + 2 = 0$，$y = \dfrac{x^2 + 2}{x} = x + \dfrac{2}{x}$。又 $y \le 4$，所以 $x + \dfrac{2}{x} \le 4$。解得 $2 - \sqrt{2} \le x \le 2 + \sqrt{2}$。
设 $\triangle AMN$ 的面积为 S，则

$$\begin{aligned} 4S^2 &= (4 + x^2) \cdot [4 + (y - x)^2] \\ &= (4 + x^2) \cdot \left(4 + \frac{4}{x^2}\right) \\ &= 20 + 4x^2 + \frac{16}{x^2} \\ &= 20 + 4\left(x^2 + \frac{4}{x^2}\right) \end{aligned}$$

这个函数在 $[2 - \sqrt{2}, \sqrt{2}]$ 上单调递减，在 $[\sqrt{2}, 2 + \sqrt{2}]$ 上单调递增，最大值是 $20 + 4 \times 12 = 68$
所以 $S_{\max} = \sqrt{17}$。

7. **Solution**. $1 + \sqrt{6}$
$\dfrac{2\sin\theta + 4\cos\theta + 5}{\sin\theta + \cos\theta + 1} = 2 + \dfrac{2\cos\theta + 3}{\sin\theta + \cos\theta + 1}$
令 $f(\theta) = \dfrac{2\cos\theta + 3}{\sin\theta + \cos\theta + 1}, f'(\theta) = \dfrac{\sin\theta - 3\cos\theta - 2}{(\sin\theta + \cos\theta + 1)^2}$，可以画出导函数分子的图像:



易知当 $\sin\theta - 3\cos\theta - 2 = 0, \theta \in (\frac{\pi}{2}, \pi)$ 取最小值，容易解得此时 $\cos\theta = \dfrac{-6 + \sqrt{6}}{10}, \sin\theta = \dfrac{2 + 3\sqrt{6}}{10}$，
故 $\left(\dfrac{2\sin\theta + 4\cos\theta + 5}{\sin\theta + \cos\theta + 1}\right)_{\min} = 1 + \sqrt{6}$

2 证明题 (本题 10 分)

Proof. 注意到：

$$\begin{aligned}|f(a)-f(b)| &= \left|\sum_{i=0}^{n-1} f\left(a+i\frac{b-a}{n}\right)-f\left(a+(i+1)\frac{b-a}{n}\right)\right| \\ &\leq \sum_{i=0}^{n-1}\left|f\left(a+i\frac{b-a}{n}\right)-f\left(a+(i+1)\frac{b-a}{n}\right)\right| \\ &\leq \sum_{i=0}^{n-1}\left|a+i\frac{b-a}{n}-\left(a+(i+1)\frac{b-a}{n}\right)\right|^2 \\ &= \frac{|a-b|^2}{n}.\end{aligned}$$

因此所证不等式成立.

3 证明题 (本题 13 分)

Proof. 我们知道函数 $f(x)=\dfrac{x}{1+x}$ 在 $[0,+\infty)$ 上递增. 则

$$\frac{|x+y|}{1+|x+y|} \leq \frac{|x|+|y|}{1+|x|+|y|} = \frac{|x|}{1+|x|+|y|}+\frac{|y|}{1+|x|+|y|} \leq \frac{|x|}{1+|x|}+\frac{|y|}{1+|y|},$$

于是

$$\frac{|x_1-x_3|}{1+|x_1-x_3|} = \frac{|x_1-x_2+x_2-x_3|}{1+|x_1-x_2+x_2-x_3|} \leq \frac{|x_1-x_2|}{1+|x_1-x_2|}+\frac{|x_2-x_3|}{1+|x_2-x_3|}$$

$$\frac{|y_1-y_3|}{1+|y_1-y_3|} = \frac{|y_1-y_2+y_2-y_3|}{1+|y_1-y_2+y_2-y_3|} \leq \frac{|y_1-y_2|}{1+|y_1-y_2|}+\frac{|y_2-y_3|}{1+|y_2-y_3|}$$

于是

$$\begin{aligned}&\max\left\{\frac{|x_1-x_3|}{1+|x_1-x_3|},\frac{|y_1-y_3|}{1+|y_1-y_3|}\right\} \\ \leq&\max\left\{\frac{|x_1-x_2|}{1+|x_1-x_2|}+\frac{|x_2-x_3|}{1+|x_2-x_3|},\frac{|y_1-y_2|}{1+|y_1-y_2|}+\frac{|y_2-y_3|}{1+|y_2-y_3|}\right\} \\ \leq&\max\left\{\frac{|x_1-x_2|}{1+|x_1-x_2|},\frac{|y_1-y_2|}{1+|y_1-y_2|}\right\} \\ &+\max\left\{\frac{|x_2-x_3|}{1+|x_2-x_3|},\frac{|y_2-y_3|}{1+|y_2-y_3|}\right\}\end{aligned}$$

即

$$|P_1P_3|_t \leq |P_1P_2|_t+|P_2P_3|_t\,.$$

注：本题出自合肥二模 2024

CHAPTER 5

2023 年启明考试数学部分

1 填空题 (每小题 9 分，共 72 分)

1. 若 α, β 为正整数且 $\alpha^2+\beta^2=3025$ 则 $\alpha+\beta=$__________.
2. $f(x)+f\left(\dfrac{1}{\sqrt[3]{1-x^3}}\right)=x^3$. 则 $f(-1)=$__________.
3. $\lim\limits_{n\to+\infty}\sin^2\sqrt{n^2+n}\pi$ 的值为__________.
4. β 为锐角. 则 $\left(1+\dfrac{1}{\sin\beta}\right)\left(1+\dfrac{1}{\cos\beta}\right)$ 最小值为__________.
5. 椭圆与双曲线
$$\frac{x^2}{m}+\frac{y^2}{p}=1,\qquad \frac{x^2}{n}-\frac{y^2}{p}=1\quad(m,n,p>0)$$
有公共焦点 F_1, F_2 和公共交点 Q，则 $S_{\triangle QF_1F_2}=$__________.
6. 设 ε 为 1 的 7 次方根 $\dfrac{\varepsilon}{1+\varepsilon^2}+\dfrac{\varepsilon^2}{1+\varepsilon^4}+\dfrac{\varepsilon^3}{1+\varepsilon^6}=$__________.
7. 三棱柱侧棱垂直底面且所有棱长为 β, 则外接球表面积为__________.
8. 甲组 5 名男同学、3 名女同学，乙组 6 名男同学、2 名女同学，从甲、乙两组各选 2 名, 则 4 人中恰好有一女的选法有__________.

2 解答题 (本题 12 分)

设数列 $\{\beta_n\}$ 满足 $\beta_1=2$，$\beta_{n+1}-\beta_n=3\times2^{n-1}$，记 $\alpha_n=n\beta_n$，求 $\{\alpha_n\}$ 的前 n 项和.

3 解答题 (本题 16 分)

已知 $f(x)=\ln(x+1)+\dfrac{\beta}{x+2}$.

(a) 若 $x>0, f(x)>1$ 恒成立，求 β 取值范围.(6 分)

(b) 求证 $\ln(2024)>\dfrac{1}{3}+\dfrac{1}{5}+\dfrac{1}{7}+\cdots+\dfrac{1}{4045}+\dfrac{1}{4047}$.(10 分)

CHAPTER 6

2023 年启明考试数学部分参考答案

1 填空题 (每小题 9 分，共 72 分)

1. **Solution**. 77

   注意到 $3025 = 55^2 = (5\times 11)^2$, 由勾股定理显然可知一种情况 $\alpha = 3\times 11, \beta = 4\times 11$, 故 $\alpha+\beta = 77$.

2. **Solution**. $\frac{1}{4}$

   注意到 $\begin{cases} f(-1) + f\left(\frac{1}{\sqrt[3]{2}}\right) = -1 \quad (1) \\ f\left(\frac{1}{\sqrt[3]{2}}\right) + f\left(\sqrt[3]{2}\right) = \frac{1}{2} \quad (2) \\ f\left(\sqrt[3]{2}\right) + f(-1) = 2 \quad (3) \end{cases}$

   $(1)+(3)-(2) \Rightarrow f(-1) = \frac{1}{4}$

3. **Solution**. 1

   $\sin^2\sqrt{n^2+n}\pi = \sin^2\left(\sqrt{n^2+n}-n\right)\pi = \sin^2\frac{n}{\sqrt{n^2+n}+n}\pi$

   $= \sin^2\frac{1}{\sqrt{1+\frac{1}{n}}+1}\pi \to \sin\frac{\pi}{2} = 1\,(n\to\infty)$

4. **Solution**. $3+2\sqrt{2}$

   $$\begin{aligned}\left(1+\frac{1}{\sin\beta}\right)\left(1+\frac{1}{\cos\beta}\right) &= \frac{\sin\beta\cos\beta+\sin\beta+\cos\beta+1}{\sin\beta\cos\beta} \\ &= \frac{\sin\beta+\cos\beta+1}{\sin\beta\cos\beta}+1\end{aligned}$$

   令 $t = \sin\beta+\cos\beta = \sqrt{2}\sin\left(\beta+\frac{\pi}{4}\right) \in \left(1, \sqrt{2}\right]$, 则 $\sin\beta\cos\beta = \frac{t^2-1}{2}$

   原式 $= 1 + \frac{1+t}{\frac{t^2-1}{2}} = 1 + \frac{2}{t-1} \geq 1 + \frac{2}{\sqrt{2}-1} = 3+2\sqrt{2}, \beta = \frac{\pi}{4}$ 取等号.

5. **Solution**. p

   由题意得 $m-n = 2p, m+n = 2(m-p)$.

   $\begin{cases} \frac{x^2}{m}+\frac{y^2}{p} = 1, \\ \frac{x^2}{n}-\frac{y^2}{p} = 1, \end{cases} \Rightarrow x^2 = \frac{2mn}{m+n}, y = \pm\frac{\sqrt{2}p}{\sqrt{m+n}}$

   $\because |F_1F_2| = 2\sqrt{m-p} = 2\sqrt{n+p}$,

   $\therefore S_{\triangle QF_1F_2} = \frac{1}{2}\times 2\sqrt{m-p}\times\frac{\sqrt{2}p}{\sqrt{m+n}} = p$.

6. **Solution**. $\dfrac{3}{2}$ or -2
对于方程 $x^7=1 \Rightarrow (x-1)\left(x^6+x^5+x^4+x^3+x^2+x+1\right)=0, \varepsilon$ 为其一个根
$\varepsilon=1$，易得原式 $=\dfrac{3}{2}$
$\varepsilon \neq 1 \Rightarrow \varepsilon^6+\varepsilon^5+\varepsilon^4+\varepsilon^3+\varepsilon^2+\varepsilon+1=0$
$$\frac{\varepsilon}{1+\varepsilon^2}+\frac{\varepsilon^2}{1+\varepsilon^4}+\frac{\varepsilon^3}{1+\varepsilon^6}=\frac{\varepsilon}{1+\varepsilon^2}+\frac{\varepsilon^5}{1+\varepsilon^3}+\frac{\varepsilon^4}{1+\varepsilon}=2\frac{\varepsilon^6+\varepsilon^4+\varepsilon}{(1+\varepsilon^2)(1+\varepsilon^3)}$$
$$=-2\frac{\varepsilon^5+\varepsilon^3+\varepsilon^2+1}{(1+\varepsilon^2)(1+\varepsilon^3)}=-2\frac{\left(1+\varepsilon^2\right)\left(1+\varepsilon^3\right)}{(1+\varepsilon^2)(1+\varepsilon^3)}=-2$$

7. **Solution**. $\dfrac{7}{3}\pi\beta^2$
由直三棱柱的几何关系可得 $R^2=\left(\dfrac{\sqrt{3}}{3}\beta\right)^2+\left(\dfrac{1}{2}\beta\right)^2 \Rightarrow R=\dfrac{\sqrt{21}}{6}\beta$. 故 $S=4\pi R^2=\dfrac{7}{3}\pi\beta^2$

8. **Solution**. 345
分两种情况：
   - 情形一: 这名女生来自甲组，共 $C_3^1C_5^1C_6^2=225$ 种
   - 情形二: 这名女生来自乙组，共 $C_2^1C_6^1C_5^2=120$ 种

故总的选法有 345 种

2 解答题 (本题 12 分)

Solution. 由错位相减法易求得 $T_n=\displaystyle\sum_{k=1}^{n}\alpha_k=3(n-1)2^n-\frac{n(n+1)}{2}+3$

3 解答题 (本题 16 分)

Solution.

(a) $\ln(x+1)+\dfrac{\beta}{x+2}>1 \Rightarrow \beta>(x+2)\left[1-\ln(x+1)\right]$
令 $h(x)=(x+2)\left[1-\ln(x+1)\right], x\in(0,+\infty)$
$h'(x)=-\left[\ln(x+1)+\dfrac{1}{x+1}\right]<0 \Rightarrow h(x)$ 在 $(0,+\infty)$ 上单调减
故 $\beta \geq h(0)=2$

(b) 由 (i) 知：$\ln(x+1)>1-\dfrac{2}{x+2}=\dfrac{x}{x+2}, x>0$
取 $x=\dfrac{1}{k}$ 得到 $\ln\left(\dfrac{k+1}{k}\right)>\dfrac{\frac{1}{k}}{\frac{1}{k}+2}=\dfrac{1}{2k+1}, x>0$
故 $\ln(2024)=\displaystyle\sum_{k=1}^{2023}\ln\left(\frac{k+1}{k}\right)>\sum_{k=1}^{2023}\frac{1}{2k+1}$

CHAPTER 7

2022 年启明考试数学部分

1 填空题 (每小题 9 分，共 72 分)

1. 已知 $\beta=\dfrac{\sqrt{5}+1}{2}$，计算 $\left[\beta^{12}\right]$（$[x]$ 为 x 的整数部分）= __________.
2. 已知: $\beta_1=3, \beta_{n+1}=\beta_n^2-3\beta_n+4$, 计算: $\displaystyle\sum_{i=1}^{\infty}\frac{1}{\beta_i-1}$__________.
3. 已知: $a^2+b^2=5$, 求 $2a+3b$ 的最大值__________.
4. 已知 $\begin{cases} x+\sin x\cos x-1=0, \\ 2\cos y-2y+\pi+4=0, \end{cases}$ 求 $\sin(2x-y)=$ __________.
5. 已知 $f(1)=2022$，且对 $n>1$ 有 $\displaystyle\sum_{k=1}^{n} f(k)=n^2 f(n)$，则 $f(2022)=$ __________.
6. 定义 $\mathbb{C}$ 为复数集，集合 $A=\{z \mid z^{18}=1, z\in\mathbb{C}\}$，$B=\{w \mid w^{48}=1, w\in\mathbb{C}\}$，求 $\operatorname{card}\{zw \mid z\in A, w\in B\}$__________.
7. 在一个凸四边形 $ABCD$ 中存在点 P，满足 $S_{\triangle PAB}=S_{\triangle PBC}=S_{\triangle PCD}=S_{\triangle PDA}$，$S_{\triangle ABC}=\alpha S_{\triangle ADC}$，求 α[1]__________.
8. 三位数中任意两个数字之和都能被第三个数整除, 求这样的三位数的个数__________.

2 解答题 (12 分)

请用 3 种方法证明: $\left(\dfrac{\alpha+\beta+\gamma}{3}\right)^3 \geq \alpha\cdot\beta\cdot\gamma\left(\alpha,\beta,\gamma\in\mathbb{R}^+\right)$.

3 解答题 (16 分)

已知: $f(x)=\beta x-\ln x-1$,

(a) 若 $f(x)\geq 0$ 恒成立, 求 β 的最小值;

(b) 求证: $\dfrac{1}{xe^x}+x+\ln x\geq 1$;

(c) 若 $\alpha\left(e^{-x}+x^2\right)\geq x-x\ln x$ 恒成立, 求 α 的取值范围

[1] 现有题面条件不足以唯一确定 α，参考答案中给出说明.

CHAPTER 8

2022 年启明考试数学部分参考答案

1 填空题 (每小题 9 分，共 72 分)

1. **Solution**. 321

   $\beta=\dfrac{\sqrt{5}+1}{2}$ 是方程 $x^2-x-1=0$ 的根, 故: $x^2=x+1, x^4=(x+1)^2=x^2+2x+1=3x+2$, $x^{12}=(3x+2)^3=27x^3+54x^2+36x+8=27x(x+1)+54(x+1)+36x+8=27x^2+117x+62$ $=27(x+1)+117x+62=144x+89\Rightarrow\beta^{12}=72\sqrt{5}+161\in(321,322)$, 故: $\left[\beta^{12}\right]=321$.

   注释: 考察对偶结构 $\left(\dfrac{\sqrt{5}+1}{2}\right)^n-\left(\dfrac{1-\sqrt{5}}{2}\right)^n$ 也是一个重要思想,

   分析: 熟知 Fibonacci i 数列通项: $a_n=\dfrac{1}{\sqrt{5}}\left[\left(\dfrac{\sqrt{5}+1}{2}\right)^n-\left(\dfrac{1-\sqrt{5}}{2}\right)^n\right]$,

   $$\left(\begin{array}{l}a_{n+2}=a_{n+1}+a_n, a_1=a_2=1, \text{则特征方程: } x^2=x+1, x_1=\dfrac{\sqrt{5}+1}{2}, x_2=\dfrac{1-\sqrt{5}}{2},\\ \text{则}: a_n=C_1\left(\dfrac{\sqrt{5}+1}{2}\right)^n+C_2\left(\dfrac{1-\sqrt{5}}{2}\right)^n,\\ \text{代入初值条件得: } a_n=\dfrac{1}{\sqrt{5}}\left[\left(\dfrac{\sqrt{5}+1}{2}\right)^n-\left(\dfrac{1-\sqrt{5}}{2}\right)^n\right]\end{array}\right),$$

   令 $a_n=\dfrac{1}{\sqrt{5}}\left[\left(\dfrac{\sqrt{5}+1}{2}\right)^n-\left(\dfrac{1-\sqrt{5}}{2}\right)^n\right]$, 则:

   $a_{12}=144=\dfrac{1}{\sqrt{5}}\left[\left(\dfrac{\sqrt{5}+1}{2}\right)^{12}-\left(\dfrac{1-\sqrt{5}}{2}\right)^{12}\right]$, $\left(\dfrac{\sqrt{5}+1}{2}\right)^{12}=144\sqrt{5}+\left(\dfrac{1-\sqrt{5}}{2}\right)^{12}$, 注意到:

   $144\sqrt{5}=\sqrt{103680}\in(321.5,322)$, $\left(\dfrac{1-\sqrt{5}}{2}\right)^{12}\approx(0.618)^{12}\approx\dfrac{1}{2^{12}}$, 故: $\left[\beta^{12}\right]=321$.

2. **Solution**. 1

   $\beta_{n+1}=\beta_n^2-3\beta_n+4\Rightarrow\beta_{n+1}-2=\beta_n^2-3\beta_n+2=(\beta_n-2)(\beta_n-1)$,

   于是

   $$\begin{aligned}\frac{1}{\beta_{n+1}-2}&=\frac{1}{(\beta_n-2)(\beta_n-1)}\\&=\frac{1}{\beta_n-2}-\frac{1}{\beta_n-1},\end{aligned}$$

   即

   $$\frac{1}{\beta_n-1}=\frac{1}{\beta_n-2}-\frac{1}{\beta_{n+1}-2},\qquad\sum_{i=1}^{\infty}\frac{1}{\beta_i-1}=1-\lim_{n\to\infty}\frac{1}{\beta_{n+1}-2}.$$

   下证: β_n 单调递增趋于 $+\infty$, 注意到: $\beta_{n+1}-\beta_n=\beta_n^2-4\beta_n+4=(\beta_n-2)^2>0$, $\Rightarrow\beta_{n+1}-\beta_1=$

$$\sum_{i=1}^{n}(\beta_i-2)^2>\sum_{i=1}^{n}(\beta_1-2)^2=n, \text{故: } \lim_{n\to\infty}\beta_n=+\infty, \text{即: } \sum_{i=1}^{\infty}\frac{1}{\beta_i-1}=1\,.$$

3. **Solution**. $\sqrt{65}$

(法一) $2a+3b\le\sqrt{2^2+3^2}\cdot\sqrt{a^2+b^2}(cauchy)=\sqrt{65}$

(法一) $2a+3b=2\sqrt{5}\cos\theta+3\sqrt{5}\sin\theta\le\sqrt{65}.$

4. **Solution**. -1

$$\begin{cases} x+\sin x\cos x-1=0 \\ 2\cos y-2y+\pi+4=0 \end{cases} \Rightarrow \begin{cases} 2x+\sin 2x-2=0 \\ y-\cos y-\dfrac{\pi}{2}-2=0 \end{cases} \Rightarrow \begin{cases} 2x+\sin 2x-2=0 \\ y-\dfrac{\pi}{2}+\sin(y-\dfrac{\pi}{2})-2=0 \end{cases}$$

, 观察上式易得同构: $2x=y-\dfrac{\pi}{2}$, 下证仅有此唯一同构
注意到: $h'(x)=(2x+\sin 2x-2=0)'=2+2\cos 2x\ge 0$,$h(x)$ 单调增
即根是唯一的，因此 $2x=y-\dfrac{\pi}{2}$ 是唯一解
故: $\sin(2x-y)=-1$.

5. **Solution**. $\dfrac{2}{2023}$

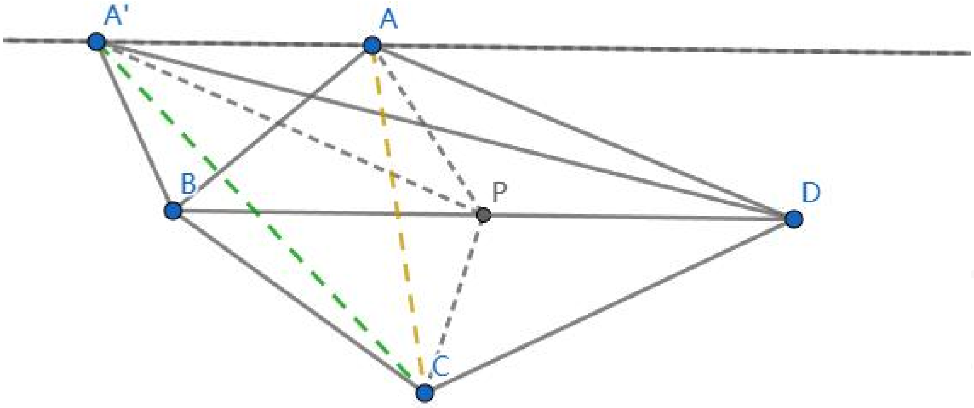
$$\begin{cases} \sum\limits_{k=1}^{n}f(k)=n^2f(n) \\ \sum\limits_{k=1}^{n+1}f(k)=(n+1)^2f(n+1) \end{cases} \Rightarrow f(n+1)=(n+1)^2f(n+1)-n^2f(n),$$

即: $\left(n^2+2n\right)f(n+1)=n^2f(n)\Rightarrow\dfrac{f(n+1)}{f(n)}=\dfrac{n}{n+2}$,

故: $f(2022)=f(1)\prod\limits_{k=1}^{2021}\dfrac{k}{k+2}=2022\cdot\dfrac{2}{2022\cdot 2023}=\dfrac{2}{2023}$.

6. **Solution**. 144
由棣莫弗定理: $z=e^{\frac{2k\pi}{18}i}=\cos\dfrac{2k\pi}{18}+i\sin\dfrac{2k\pi}{18}, w=e^{\frac{2k'\pi}{48}i}=\cos\dfrac{2k'\pi}{48}+i\sin\dfrac{2k'\pi}{48}$,
$zw=e^{\frac{2(8k+3k')}{144}i\pi}$, 注意到 8 与 3 互质, 则根据裴蜀定理, $8k+3k'$ 可以取得任意整数值, $k,k'\in\mathbb{Z}$
$8k+3k'\in\mathbb{Z}$, 且 $8k+3k'$ 可以取到 $[0,143]$ 的任意值, 故: zw 一共有 144 个.

7. **Solution**. 现有题面无法唯一确定 α.
取 BD 为四边形 $ABCD$ 的一条面积平分线，并令 P 为 BD 的中点，则 $S_{\triangle PAB}=S_{\triangle PDA}$ 且 $S_{\triangle PBC}=S_{\triangle PCD}$；又因 BD 平分四边形面积，四个面积彼此相等. 保持点 B,C,D 不动，使点 A 沿平行于 BD 的直线移动，上述四个小三角形面积仍相等，但 $S_{\triangle ABC}/S_{\triangle ADC}$ 会发生变化. 因此还需要补充条件才能确定 α.



8. **Solution**. 39

分三种情况:

- 情形一: 该三位数各位数字相等, 显然有 9 个,
- 情形二: 该三位数各位数字不等, 有: $(1,2,3),(2,4,6),(3,6,9)$, 一共 $3A_3^3=18$ 个,
- 情形三: 该三位数有且仅有 2 位数相同, 有: $(1,1,2),(2,2,4),(3,3,6),(4,4,8)$, 一共 $4C_3^1=12$ 个,

故这样的三位数有 : $9+18+12=39$ 个.

2 解答题 (12 分)

Solution.

(法一) 根据基本不等式 $\dfrac{\alpha+\beta}{2}\geq\sqrt{\alpha\beta}$, 故:

$$\frac{\alpha+\beta+\gamma+\eta}{4}=\frac{\frac{\alpha+\beta}{2}+\frac{\gamma+\eta}{2}}{2}\geq\sqrt{\sqrt{\alpha\beta}\sqrt{\gamma\eta}}=\sqrt[4]{\alpha\cdot\beta\cdot\gamma\cdot\eta}$$

令 $\eta=\dfrac{\alpha+\beta+\gamma}{3}$, 有: $\dfrac{\alpha+\beta+\gamma}{3}\geq\sqrt[3]{\alpha\cdot\beta\cdot\gamma}$, 证毕

(法二) 令 $\alpha=x^3,\beta=y^3,\gamma=z^3$, 则:

$$\begin{aligned}\frac{x^3+y^3+z^3}{3}-xyz=&\frac{1}{3}\left(x^3+y^3+z^3-3xyz\right)\\=&\frac{1}{3}\left\{\left[(x+y)^3-3x^2y-3xy^2\right]+z^3-3xyz\right\}\\=&\frac{1}{3}\left\{\left[(x+y)^3+z^3\right]-3x^2y-3xy^2-3xyz\right\}\\=&\frac{1}{6}(x+y+z)\left[(x-y)^2+(y-z)^2+(z-x)^2\right]\geq0\end{aligned}$$

证毕

(法三) 令 $f(x)=\ln x$, 则: $f'(x)=\dfrac{1}{x},f''(x)=-\dfrac{1}{x^2}<0$,

故: $f(x)$ 是上凸函数, 根据琴生不等式, $\ln\dfrac{\alpha+\beta+\gamma}{3}\geq\dfrac{1}{3}(\ln\alpha+\ln\beta+\ln\gamma)$, 证毕

3 解答题 (16 分)

Solution.

(a) $f(x)$ 的定义域是 $(0,+\infty)$, $f(x)\geq 0 \Rightarrow \beta x-\ln x-1\geq 0, \Rightarrow \beta\geq\dfrac{\ln x+1}{x}$,
令 $g(x)=\dfrac{\ln x+1}{x}, g'(x)=-\dfrac{\ln x}{x^2}$,
当 $x\in(0,1)$ 时, $g'(x)>0, g(x)$ 单调增, 当 $x\in(1,+\infty)$ 时, $g'(x)<0, g(x)$ 单调减
故 $g_{\max}(x)=g(1)=1$, 即: $\beta\geq g_{\max}(x)=1, \beta$ 的最小值为 1.

(b) 证明: 当 $\beta=1$ 时, 有: $\ln x\leq x-1$,
令 $\dfrac{1}{xe^x}=t$, 则 : $-x-\ln x=\ln t\leq t-1=\dfrac{1}{xe^x}-1$, 即: $\dfrac{1}{xe^x}+x+\ln x\geq 1$.

(c) : $\alpha\left(e^{-x}+x^2\right)\geq x-x\ln x\Rightarrow\alpha\left(\dfrac{1}{xe^x}+x\right)\geq 1-\ln x$,
注意到: $\dfrac{1}{xe^x}+x>0$, 即: $\alpha\geq\dfrac{1-\ln x}{\dfrac{1}{xe^x}+x}$,
由 (b) 知: $\dfrac{1-\ln x}{\frac{1}{xe^x}+x}\leq 1$, 且 $xe^x=1$ 时, $\dfrac{1-\ln x}{\dfrac{1}{xe^x}+x}=1$,
故 : $\alpha\geq 1, \alpha$ 的取值范围是 $[1,+\infty)$.

CHAPTER 9

2021 年启明考试数学部分

1　填空题

1. $x, y, z \in \mathbb{Q}, \sqrt{3+\sqrt{2}+\sqrt{3}+\sqrt{6}} = \sqrt{x}+\sqrt{y}+\sqrt{z}, xyz =$ __________.
2. $f(x) \geq 0, f^2(x+1)+f^2(x) = 73(x \in \mathbb{R})$. 当 $x \in [0,1)$ 时,
   $f(x) = 10-|13x-4|, f(\dfrac{2021}{13}) =$__________.
3. $y = \dfrac{4\sin\theta\cos\theta+3}{\sin\theta+\cos\theta}(-\dfrac{\pi}{4}<\theta<\dfrac{3\pi}{4})$, 则 y 的最小值为__________.
4. $\{a_n\}$ 是等差数列, 且 $a_3=-13, a_7=3. S_n=\sum\limits_{i=1}^{n} a_n$, 则 $(S_n)_{min} =$__________.
5. $w = \cos\dfrac{\pi}{5}+i\sin\dfrac{\pi}{5}$, 则以 w, w^3, w^7, w^9 为解的方程为__________.
6. A, B 是椭圆 $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1(a>b>0)$ 上两点，O 是原点，OA⊥OB，则 $|AB|_{min} =$__________.
7. 双曲线 $\Gamma: \dfrac{x^2}{4}-\dfrac{y^2}{5}=1$，过右焦点作一条长度为 $4\sqrt{3}$ 的弦 AB，且 A, B 位于右支上. 将双曲线 Γ 绕其右准线旋转 90°，则弦 AB 形成的曲面面积为__________.

2　解答题

三种方法求 $x(x+1)(x+2)(x+3)$ 最小值.

3　证明题

已知 $0<a<b$,证 $\dfrac{\ln b-\ln a}{b-a} > \dfrac{2a}{a^2+b^2}$.

CHAPTER 10

2021 年启明考试数学部分参考答案

1 填空题

1. **Solution**. $\frac{3}{4}$.
注意到
$$\left(1+\sqrt{\frac{1}{2}}+\sqrt{\frac{3}{2}}\right)^2=3+\sqrt{2}+\sqrt{3}+\sqrt{6}.$$
因此可取
$$x=1,\qquad y=\frac{1}{2},\qquad z=\frac{3}{2},$$
从而 $xyz=\frac{3}{4}$.

2. **Solution**. $f^2(x+1)+f^2(x)=73$
$\Rightarrow f^2(x+2)+f^2(x+1)=73\Rightarrow f^2(x+2)=f^2(x)$
$\because f(x)\geq 0\therefore f(x+2)=f(x)$
$f(\frac{2021}{13})=f(\frac{19}{13})$, 又 $\because f^2(\frac{19}{13})+f^2(\frac{6}{13})=73$
解得:$f(\frac{2021}{13})=3$

3. **Solution**. $y=\frac{2(1+2\sin\theta\cos\theta)+1}{\sin\theta+\cos\theta}=\frac{2(\sin\theta+\cos\theta)^2+1}{\sin\theta+\cos\theta}$
$=2(\sin\theta+\cos\theta)+\frac{1}{\sin\theta+\cos\theta}\geq 2\sqrt{2}$
当且仅当 $2(\sin\theta+\cos\theta)=\frac{1}{\sin\theta+\cos\theta}$, 即 $\theta=\frac{7\pi}{12}or-\frac{\pi}{12}$ 取 ” = ”

4. **Solution**. $\begin{cases}a_1+2d=-13\\a_1+6d=3\end{cases}\Rightarrow\begin{cases}a_1=-21\\d=4\end{cases}$
易知 $n=6,(S_n)_{\min}=-66$

5. **Solution**. $x^4-x^3+x^2-x+1=0$.
因为 $w=\mathrm{e}^{\pi\mathrm{i}/5}$，所以 w,w^3,w^7,w^9 恰为全部 10 次本原单位根. 它们是第 10 个分圆多项式
$$\Phi_{10}(x)=\frac{x^5+1}{x+1}=x^4-x^3+x^2-x+1$$
的四个根.

6. **Solution**. 根据题意不妨设 $A\left(r_1\cos\theta,r_1\sin\theta\right),B\left[r_2\cos\left(\theta+\frac{\pi}{2}\right),r_2\sin\left(\theta+\frac{\pi}{2}\right)\right]$,
则 $B\left(-r_2\sin\theta,r_2\cos\theta\right)$, $AB^2=r_1^2+r_2^2$, $\frac{r_1^2\cos^2\theta}{a^2}+\frac{r_1^2\sin^2\theta}{b^2}=1$,

$$r_1^2=\frac{a^2b^2}{b^2\cos^2\theta+a^2\sin^2\theta},\qquad r_2^2=\frac{a^2b^2}{b^2\sin^2\theta+a^2\cos^2\theta}.$$

$$\begin{aligned}
\text{故}r_1^2+r_2^2&=\frac{a^2b^2\left(a^2+b^2\right)}{\left(b^2\cos^2\theta+a^2\sin^2\theta\right)\left(b^2\sin^2\theta+a^2\cos^2\theta\right)}\\
&=\frac{a^2b^2\left(a^2+b^2\right)}{\left(a^4+b^4\right)\sin^2\theta\cdot\cos^2\theta+a^2b^2\left(\sin^4\theta+\cos^4\theta\right)}\\
&=\frac{a^2b^2\left(a^2+b^2\right)}{\left(a^4+b^4\right)\sin^2\theta\cdot\cos^2\theta+a^2b^2\left(1-2\sin^2\theta\cdot\cos^2\theta\right)}\\
&=\frac{a^2b^2\left(a^2+b^2\right)}{\left(a^2-b^2\right)^2\sin^2\theta\cdot\cos^2\theta+a^2b^2}=\frac{4a^2b^2\left(a^2+b^2\right)}{\left(a^2-b^2\right)^2\sin^2 2\theta+4a^2b^2}\\
&\geqslant\frac{4a^2b^2\left(a^2+b^2\right)}{\left(a^2+b^2\right)^2}.
\end{aligned}$$

当且仅当 $\theta=k\pi\pm\dfrac{\pi}{4}(k\in\mathbf{Z})$ 时等号成立. 因此，线段 AB 长度的最小值为 $\dfrac{2ab}{\sqrt{a^2+b^2}}$.

7. **Solution**. 分析易知弦 AB 扫过的面积为一圆台侧面积的 $\dfrac{1}{4}$. 设 A,B 两点到右准线 $l:x=\dfrac{4}{3}$ 的距离分别为 r_1,r_2,
由于双曲线离心率 $e=\dfrac{3}{2}$, 故 $S=\dfrac{1}{4}\pi(r_1+r_2)|AB|=\dfrac{1}{4e}\pi|AB|^2=8\pi$

2 解答题

Solution.

(法一) ：不妨设 $y=x+\dfrac{3}{2}$，则 $x=y-\dfrac{3}{2}$，代入 $x(x+1)(x+2)(x+3)$ 中得到：

$$\begin{aligned}
&x(x+1)(x+2)(x+3)\\
=&(y-\frac{3}{2})(y-\frac{1}{2})(y+\frac{1}{2})(y+\frac{3}{2})\\
=&(y^2-\frac{9}{4})(y^2-\frac{1}{4})\\
=&y^4-\frac{5}{2}y^2+\frac{9}{16}\\
=&(y^2-\frac{5}{4})^2-1
\end{aligned}$$

由于 $y^2\geq 0$，所以 $x(x+1)(x+2)(x+3)\geq -1$，等号成立当且仅当 $y^2=\dfrac{5}{4}$，即 $x=\dfrac{\pm\sqrt{5}-3}{2}$

(法二) ：

$$\begin{aligned}
&\frac{d}{dx}[x(x+1)(x+2)(x+3)]\\
&=(x+1)(x+2)(x+3)+x(x+2)(x+3)+x(x+1)(x+3)+x(x+1)(x+2)\\
&=4x^3+18x^2+22x+6\\
&=(4x+6)(x^2+3x+1)
\end{aligned}$$

令导函数等于 0，解得 $x=-\dfrac{3}{2}$ 或 $x=\dfrac{\pm\sqrt{5}-3}{2}$ 带入原函数中可得最小值在 $x=\dfrac{\pm\sqrt{5}-3}{2}$ 取得，结果为 -1。

(法三)：

$$
\begin{aligned}
& x(x+1)(x+2)(x+3) \\
=& x(x+3)(x+1)(x+2) \\
=& (x^2+3x)(x^2+3x+2) \\
=& (x^2+3x+1-1)(x^2+3x+1+1) \\
=& (x^2+3x+1)^2-1 \geq -1 (x^2+3x+1=0, \text{即} x=\frac{\pm\sqrt{5}-3}{2} \text{取得})
\end{aligned}
$$

3 证明题

Proof. 原式等价于要证：$\ln\frac{b}{a} > \frac{2a(b-a)}{a^2+b^2} = \frac{2(\frac{b}{a}-1)}{1+(\frac{b}{a})^2}$, 设 $t=\frac{b}{a}\in(1,+\infty)$，即证:$(1+t^2)\ln t > 2t-2$
注意到 $\ln t > 1-\frac{1}{t}, t\in(1,+\infty)$, 即证：$(1+t^2)\ln t > (1+t^2)(1-\frac{1}{t}) > 2t-2$,
而

$$t^2-3t+3-\frac{1}{t}=\frac{(t-1)^3}{t}>0,$$

故原不等式成立.

CHAPTER 11

2019 年启明考试数学部分

1 填空题

1. $37, 23, 19, 5.....$ 以此类推, 第五个数为__________.
2. 今天星期六，则 3^{1998} 天后是星期几__________.
3. 若 x 满足 $x + x^{-1} = -1$, 求 $x^{2019} + x^{-2019}$__________.
4. 求方程 $13\sin x = x$ 的根__________.
5. 设整数数列 $a_1, a_2, ...a_{10}$ 满足 $a_{10} = 3a_1, a_2 + a_8 = 2a_5$. 且 $a_{i+1} \in \{1 + a_i, 2 + a_i\}(i = 1, 2, 3...9)$, 求这样的数列有多少个__________.
6. 求 $n^3 + 100$ 能被 $n + 10$ 整除的最大整数__________.
7. x, y, z 为非负整数，且 $\begin{cases} x + y + z = 10 & ① \\ x + 2y + 3z = 30② \end{cases}$ ，求 $x + 5y + 3z$ 的取值范围__________.
8. 若 $f(x) = (1 - x^2)\left(x^2 + ax + b\right)$ 关于 $x = -2$ 对称, 求 $f(x)_{\max}$__________.
9. 若 α, β 满足 $\begin{cases} x\sin\beta + y\cos\alpha = \sin\alpha \\ x\sin\alpha + y\cos\beta = \sin\beta \end{cases}$, 求 $x^2 - y^2$__________.
10. $A(3, 2)$ 是平面上一点，F 是抛物线:$y^2 = 2x$ 上的焦点，抛物线上一点 M, 则 $|MF| + |MA|$ 取最小值的时 M 的坐标__________.

2 证明题

$\forall x, y \in \mathbb{R}$, 函数 $f(z)$ 满足 $2f(x)f(y) = f(x + y) + f(x - y)$ 且 $f(x) \neq 0$, 证明:$f^2(x) + f^2(y) = f(x + y)f(x - y) + 1$

3 解答题

数列 $\{a_n\}$ 满足 $a_1 = 3, a_{n+1} \leq \dfrac{1}{2}(a_n + a_{n+2})$, 且 $a_{505} = 2019$. 求 $\max a_5$.

4 解答题

函数 f 是 $\mathbb{R}$ 上的偶函数，其图像关于 $x=1$ 对称。对任意 $x,y\in\left[0,\frac{1}{2}\right]$，有

$$f(x+y)=f(x)f(y),\qquad f(1)=\beta>0.$$

(1) 求 $f\left(\frac{1}{2}\right),f\left(\frac{1}{4}\right)$ 的值

(2) 证明：$f(x)$ 是周期函数

(3) 记 $\beta(n)=f\left(2n+\frac{1}{2n}\right)$, 求 $\lim\limits_{n\to\infty}\ln\beta(n)$.

CHAPTER 12

2019 年启明考试数学部分参考答案

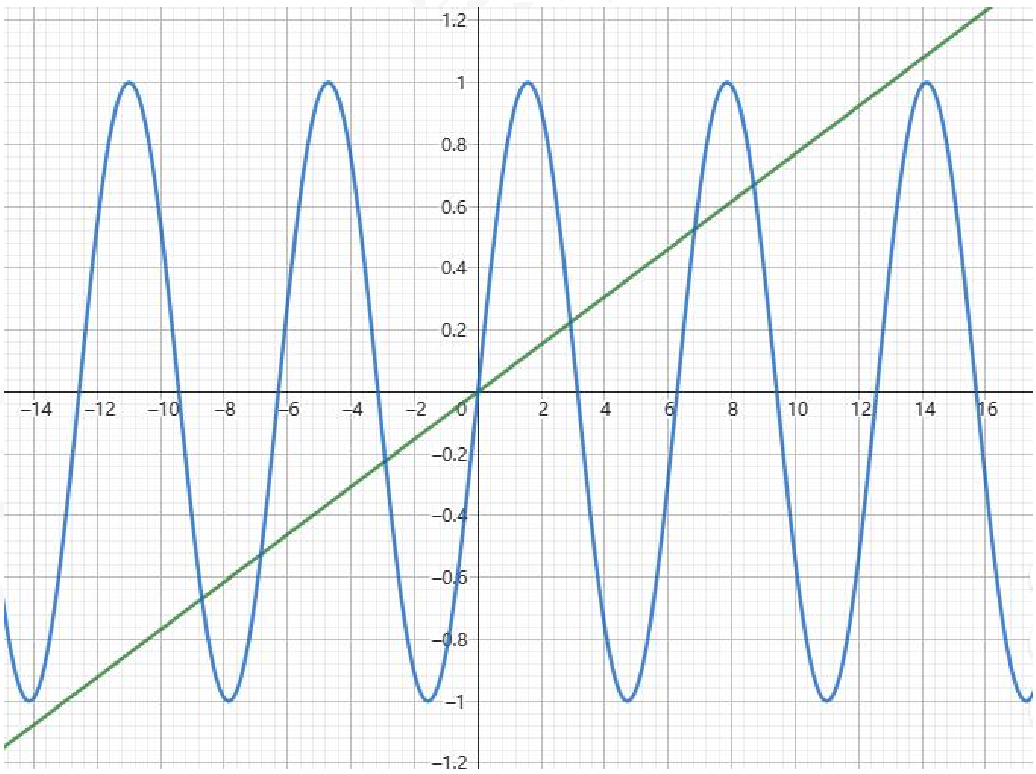
1 填空题

1. **Solution**. 1

2. **Solution**. 注意到 $3^{1998}=(27)^{666}=(28-1)^{666}=1+\sum\limits_{k=0}^{665}C_{666}^{k}(28)^{666-k}(-1)^{k}=7N+1$, 故为星期天.

3. **Solution**. $x+x^{-1}=-1\Rightarrow x^2+x+1=0$, 则 $x=-\dfrac{1}{2}\pm\dfrac{\sqrt{3}}{2}i=e^{\pm\frac{2\pi}{3}i}$(欧拉公式).
   由棣莫弗定理易得 $x^{2019}=e^{\pm1046\pi i}=1$, 故 $x^{2019}+x^{-2019}=2$.

4. **Solution**. 作图:



   可以求得的零点为 $x=0$，由于此方程为超越方程，其他 6 个零点为隐零点.

5. **Solution**. 不妨设 $b_i=a_{i+1}-a_i\in\{1,2\}$, 由 $a_{10}=3a_1, a_2+a_8=2a_5$, 可得:

$$\begin{cases} 2a_1 = \sum_{i=1}^{9} b_i \\ b_7+b_6+b_5=b_4+b_3+b_2 \end{cases}$$

对于 $b_7+b_6+b_5=b_4+b_3+b_2$, 分析易知等式两端 $1,2$ 的个数相等，因此 b_7,b_6,b_5,b_4,b_3,b_2 总的取法共有：$(C_3^0)^2+(C_3^1)^2+(C_3^2)^2+(C_3^3)^2=20$ 种

注意到 $b_7+b_6+b_5+b_4+b_3+b_2$ 必是偶数，由 $\because 2a_1=\sum_{i=1}^{9} b_i$, $2a_1$ 也为偶数

故 $b_1+b_8+b_9$ 必为偶数，取法有 4 种

故满足条件的数列共有 $4\times 20=80$ 种

6. **Solution**.
$$\begin{aligned} &\frac{n^3+100}{n+10} \\ =&\frac{n^3+1000}{n+10}-\frac{900}{n+10} \\ =&n^2-10n+100-\frac{900}{n+10} \end{aligned}$$
$\Rightarrow n_{max}=890$

7. **Solution**. $3\times ① - ② \Rightarrow 2x+y=0$
$\because x,y\in\mathbb{N}^*\ \therefore x=y=0\Rightarrow z=10$
故 $x+5y+3z=30$

8. **Solution**. 注意到 $x=-1,1$ 为 $f(x)$ 的零点，又 $\because x=-2$ 为其对称轴，因此可得另两个零点为：$x=-3,x=-5$
故 $f(x)=(1-x^2)(x+3)(x+5)$

$$\begin{aligned} \text{注意到:}f(x)&=(1-x)(x+5)(1+x)(x+3) \\ &=(-x^2-4x+5)(x^2+4x+3) \\ &\le(\frac{-x^2-4x+5+x^2+4x+3}{2})^2 \\ &=16(\text{当且仅当}-x^2-4x+5=x^2+4x+3\text{取得}) \end{aligned}$$

9. **Solution**. 分析：不要被题设吓到，实际上就是解二元一次方程组

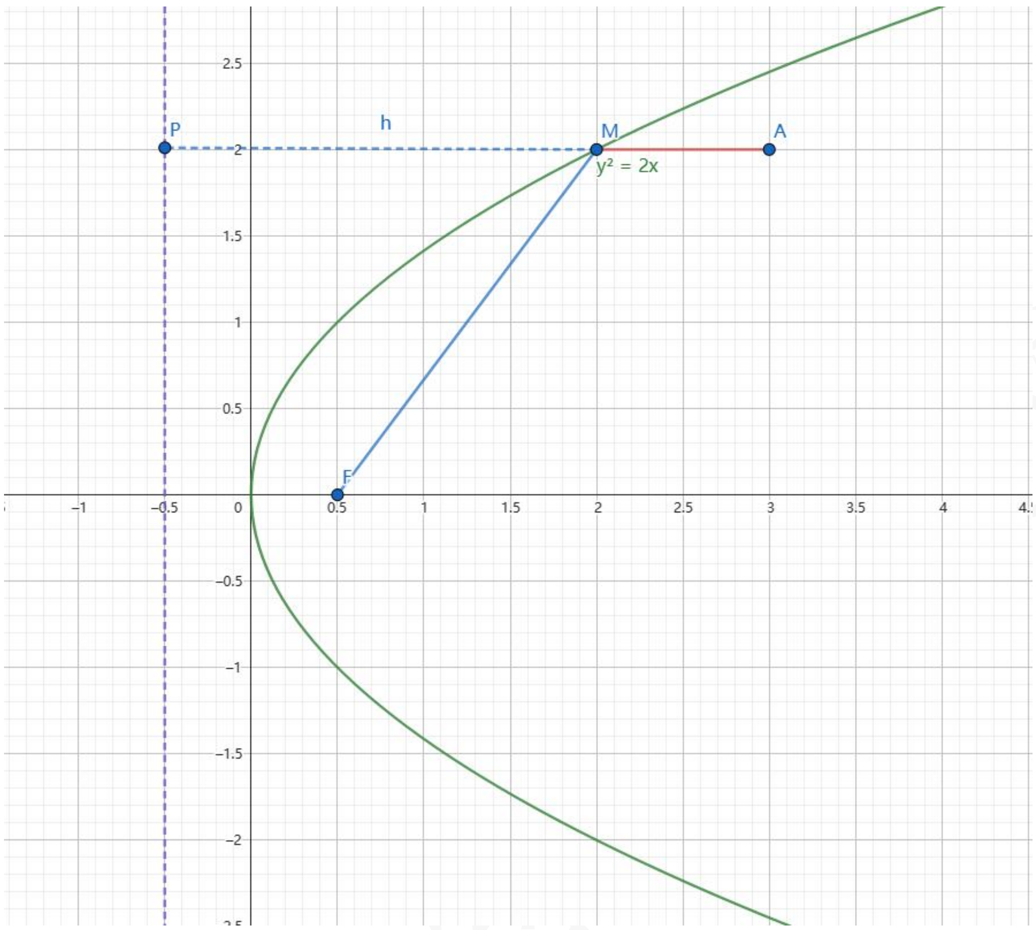
$$\begin{cases} x\sin\beta+y\cos\alpha=\sin\alpha \\ x\sin\alpha+y\cos\beta=\sin\beta \end{cases} \Rightarrow \begin{cases} x=\dfrac{\sin\beta\cos\alpha-\sin\alpha\cos\beta}{\sin\alpha\cos\alpha-\sin\beta\cos\beta} \\ y=\dfrac{\sin^2\alpha-\sin^2\beta}{\sin\alpha\cos\alpha-\sin\beta\cos\beta} \end{cases}$$

$$\begin{aligned} x&=\frac{\sin\beta\cos\alpha-\sin\alpha\cos\beta}{\sin\alpha\cos\alpha-\sin\beta\cos\beta}=\frac{2\sin(\alpha-\beta)}{\sin2\alpha-\sin2\beta} \\ &=\frac{2\sin(\alpha-\beta)}{2\cos(\alpha+\beta)\sin(\alpha-\beta)}=-\sec(\alpha+\beta) \end{aligned}$$

$$\begin{aligned} y&=\frac{\sin^2\alpha-\sin^2\beta}{\sin\alpha\cos\alpha-\sin\beta\cos\beta}=\frac{\cos2\beta-\cos2\alpha}{\sin2\alpha-\sin2\beta} \\ &=\frac{\sin(\alpha+\beta)\sin(\alpha-\beta)}{\sin(\alpha-\beta)\cos(\alpha+\beta)}=\tan(\alpha+\beta) \end{aligned}$$

故 $x^2-y^2=1$

10. **Solution**. 由题意得 $F(\frac{1}{2},0)$, 准线方程为 $x=-\frac{1}{2}$, 设点 M 到准线的距离为 $d=|PM|$. 则由抛物线的定义得 $|MA|+|MF|=|MA|+|PM|$ 故当 P,A,M 三点共线时，$|MF|+|MA|$ 取得最小值为 $|AP|=3-(-\frac{1}{2})=\frac{7}{2}$. 把 $y=2$ 代入抛物线 $y^2=2x$ 得 $x=2$，故点 M 的坐标是 $(2,2)$
附个图：



2 证明题

Proof. 令 $x=y=0 \Rightarrow f(0)=1$

$2f(x)f(y)=f(x+y)+f(x-y) \Rightarrow 2f(x+y)f(x-y)=f(2x)+f(2y)(*)$

令 $x=y \Rightarrow 2f^2(x)=f(2x)+f(0)=f(2x)+1$, 同理 $2f^2(y)=f(2y)+1$

带入 $(*)$ 中得：$f^2(x)+f^2(y)=f(x+y)f(x-y)+1$

3 解答题

Solution. 分析：一个自然的想法是所有的等号均能取得时 a_5 取得最大值，此时 $\{a_n\}$ 为 $d=4$ 的等差数列，解得 $a_5=19$

令 $b_n=a_{n+1}-a_n$,$a_{n+1} \leq \frac{1}{2}(a_n+a_{n+2}) \Rightarrow b_n \leq b_{n+1}$

$\therefore a_5=a_1+b_1+b_2+b_3+b_4$

$a_{505}-a_1=\sum_{i=1}^{504} b_i \geq 126(b_1+b_2+b_3+b_4) \Rightarrow b_1+b_2+b_3+b_4 \leq 16$

$\therefore a_5=a_1+b_1+b_2+b_3+b_4 \leq 19$, 当 $\{a_n\}$ 为等差数列取 ”$=$”

4 解答题

Solution.

(1) 易得:$f(\frac{1}{2})=\beta^{\frac{1}{2}}, f(\frac{1}{4})=\beta^{\frac{1}{4}}$

(2) **Proof**. 由偶对称与关于直线 $x=1$ 的轴对称性

$f(2+x)=f(-x)=f(x)\Rightarrow f(x)$是$T=2$的周期函数

(3) 由周期性，$\beta(n)=f\left(2n+\frac{1}{2n}\right)=f\left(\frac{1}{2n}\right)$.

注意到:$f(\frac{1}{2})=f(n\times\frac{1}{2n})=f^n(\frac{1}{2n})\Rightarrow f(\frac{1}{2n})=\beta^{\frac{1}{2n}}$

因为 $f(1/2)=\sqrt{\beta}$，所以

$$f\left(\frac{1}{2n}\right)^n=f\left(\frac{1}{2}\right)=\sqrt{\beta}.$$

各项均为正，故 $\beta(n)=\beta^{1/(2n)}$，从而

$$\lim_{n\to\infty}\ln\beta(n)=\lim_{n\to\infty}\frac{\ln\beta}{2n}=0.$$

CHAPTER 13

2015 年启明考试数学部分

1 填空题

1. 对抛物线 $y^2 = 2\sqrt{2}x$, 若设其焦点为 F, y 轴正半轴上一点为 N. 若准线上存在唯一的点 P 使得 $\angle NPF = 90°$, 则 N 点的纵坐标为__________.
2. $\dfrac{1}{\sqrt{1}+\sqrt{2}} + \dfrac{1}{\sqrt{2}+\sqrt{3}} + \cdots + \dfrac{1}{\sqrt{255}+\sqrt{256}} =$__________.
3. 若已知 $\lim\limits_{n\to\infty}\left(\sum\limits_{i=1}^{n}\dfrac{1}{i} - \ln n\right)$ 存在, 则 $\sum\limits_{i=0}^{\infty}\dfrac{(-1)^i}{i+1} =$__________.
4. 在边长为 1 的正方形中（含边界）取 9 个点，其中必有 3 个点，
   它们构成的三角形面积不超过__________.
5. 某人打靶打中 8 环、9 环、10 环的概率分别为 0.15, 0.25, 0.2，现开三枪，
   不少于 28 环的概率为__________.

2 解答题

若对任意实数 x, y, 有 $f\left((x-y)^2\right) = (f(x))^2 - 2xf(y) + y^2$, 求 $f(x)$.

3 解答题

求所有 a, b, 使 $\left|\sqrt{1-x^2} - ax - b\right| \leqslant \dfrac{\sqrt{2}-1}{2}$ 成立, 其中 $x \in [0,1]$.

4 解答题

若复数 z 满足 $|z| = 1$, 求 $\left|z^3 - z + 2\right|^2$ 的最小值.

5 解答题

已知三次方程 $x^3 + ax^2 + bx + c = 0$ 有三个实根.

(a) 若三个实根为 x_1, x_2, x_3, 且 $x_1 \leqslant x_2 \leqslant x_3, a, b$ 为常数, 求 c 变化时 $x_3 - x_1$ 的取值范围;

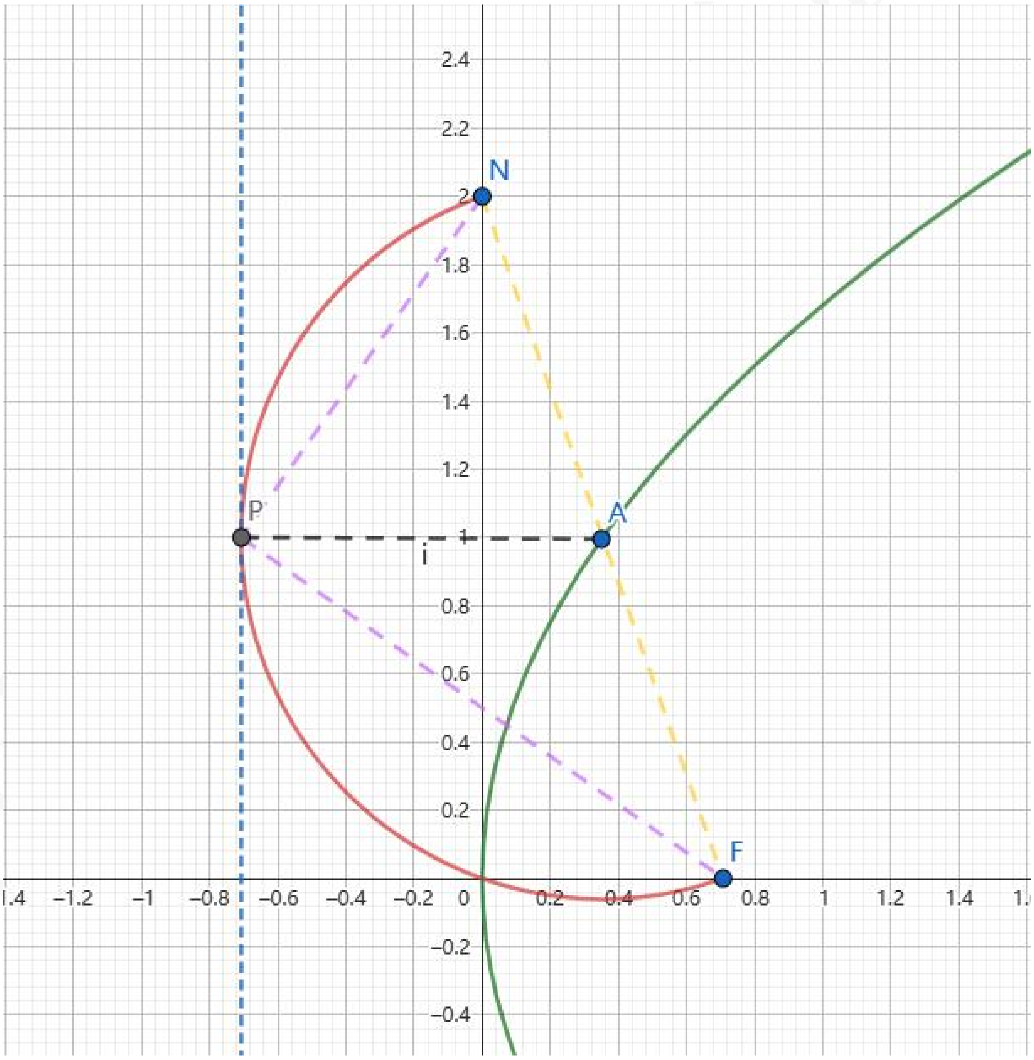
(b) 若三个实根为 a, b, c, 求 a, b, c.

CHAPTER 14

2015 年启明考试数学部分参考答案

1 填空题

1. **Solution**. 分析：实际上就是以 NF 为直径的圆与准线相切，求此时 N 点坐标，图如下：



$|AP| = |AN| = |AF|, AP \perp l : x = \dfrac{\sqrt{2}}{2} \Rightarrow A$ 在抛物线上

设 $N(0, a) \Rightarrow A(\dfrac{\sqrt{2}}{4}, \dfrac{a}{2})$ 带入抛物线方程，解得 $a = 2$

2. **Solution**. 设 $b_n = \dfrac{1}{\sqrt{n+1}+\sqrt{n}} = \sqrt{n+1}-\sqrt{n}$

$\dfrac{1}{\sqrt{1}+\sqrt{2}} + \dfrac{1}{\sqrt{2}+\sqrt{3}} + \cdots + \dfrac{1}{\sqrt{255}+\sqrt{256}} = \sum\limits_{i=1}^{255} b_i = \sqrt{256}-1 = 15$

3. **Solution**. 记调和数 $H_n = \sum\limits_{k=1}^{n} \dfrac{1}{k}$，可用以下两种方法计算

(法一) 设 $H(n) = \sum\limits_{k=1}^{n} \dfrac{1}{k} \Rightarrow H(n) = \gamma + \ln n + o(1)$

考虑部分和: $S_{2n} = \sum\limits_{i=1}^{2n} \dfrac{(-1)^{i-1}}{i} = \sum\limits_{i=1}^{n} \dfrac{1}{2i-1} - \sum\limits_{i=1}^{n} \dfrac{1}{2i} = \sum\limits_{i=1}^{2n} \dfrac{1}{i} - \sum\limits_{i=1}^{n} \dfrac{1}{i} = H(2n) - H(n) = \ln 2 + o(1) \to \ln 2 (n \to \infty)$

$S_{2n+1} = S_{2n} + \dfrac{1}{2n+1} \to \ln 2 (n \to \infty)$

综上: $\sum\limits_{i=0}^{\infty} \dfrac{(-1)^i}{i+1} = \ln 2$

(法二) 注意到: $\ln(1+x) = \sum\limits_{n=1}^{\infty} \dfrac{(-1)^{n-1}}{n} x^n \Rightarrow \ln 2 = \sum\limits_{n=1}^{\infty} \dfrac{(-1)^{n-1}}{n}$

4. **Solution**. 在边长为 1 的正方形内任取 3 个点，则这 3 点组成的三角形的面积不超过 $\dfrac{1}{2}$。若我们在一个边长为 $\dfrac{1}{2}$ 的正方形内任取 3 点, 那么这 3 点所组成的三角形的最大面积就是这个正方形的面积的一半，即不超过 $\dfrac{1}{8}$。那么我们将一个边长为 1 的正方形分为 4 个相同的边长为 $\dfrac{1}{2}$ 的小正方形，任取 9 个点放到正方形中，必有 3 个点落在同一个小正方形内。
由上面的分析得知, 这 3 个点所组成的三角形面积不超过 $\dfrac{1}{8}$

5. **Solution**.

- 情形一: 成绩为 28 环，有以下情况:$\{9,9,10\},\{8,10,10\}$
- 情形二: 成绩为 29 环，有以下情况:$\{9,10,10\}$
- 情形三: 成绩为 30 环，有以下情况:$\{10,10,10\}$

$p = \mathbf{C}_3^2 0.25^2 \cdot 0.2 + \mathbf{C}_3^2 0.2^2 \cdot 0.15 + \mathbf{C}_3^2 0.2^2 \cdot 0.25 + 0.2^3 = 0.0935$

2 解答题

Solution. $f(x) = x$ 或 $f(x) = x+1$.

令 $x = y$，得

$$f(0) = (f(x)-x)^2.$$

再令 $x = y = 0$，得 $f(0) = f(0)^2$，故 $f(0) = 0$ 或 $f(0) = 1$.

若 $f(0) = 0$, 则 $(f(x)-x)^2 = 0$, 所以 $f(x) = x$. 若 $f(0) = 1$, 则 $f(x) = x+\varepsilon(x)$, 其中 $\varepsilon(x) \in \{-1, 1\}$. 将其代回原方程，并注意到 $(x-y)^2 \geq 0$ 及 $f(t) = t+1$ $(t \geq 0)$，可得

$$2x\big(\varepsilon(x) - \varepsilon(y)\big) = 0.$$

取 $x \neq 0$ 可知 $\varepsilon(y)$ 与 y 无关，因此 $\varepsilon \equiv 1$ 或 $\varepsilon \equiv -1$. 又 $f(t) = t+1$ $(t \geq 0)$，故只能有 $\varepsilon \equiv 1$，即 $f(x) = x+1$. 直接代入可知两个函数均满足原方程.

3 解答题

Solution. $a=-1$，$b=\dfrac{1+\sqrt{2}}{2}$.

令 $x=\sin\theta$，其中 $0\le\theta\le\dfrac{\pi}{2}$，并记

$$E=\frac{\sqrt{2}-1}{2},\qquad g(\theta)=\cos\theta-a\sin\theta-b.$$

题设等价于 $|g(\theta)|\le E$ 对所有 $\theta\in[0,\pi/2]$ 成立. 令

$$e_0=g(0)=1-b,\qquad e_1=g\left(\frac{\pi}{2}\right)=-a-b.$$

则 $|e_0|,|e_1|\le E$，并且

$$g\left(\frac{\pi}{4}\right)=\sqrt{2}-1+\left(1-\frac{1}{\sqrt{2}}\right)e_0+\frac{1}{\sqrt{2}}e_1\ge 2E-E=E.$$

另一方面 $|g(\pi/4)|\le E$，所以各处必须同时取等号，即 $e_0=e_1=-E$. 由此解得

$$a=-1,\qquad b=1+E=\frac{1+\sqrt{2}}{2}.$$

反之，当 $a=-1$ 且 $b=(1+\sqrt{2})/2$ 时，$1\le\sin\theta+\cos\theta\le\sqrt{2}$，故 $|g(\theta)|\le E$.

4 解答题

Solution.

(法一) 设 $z=\cos\theta+i\sin\theta$, 得 $z^3-z+2=\cos3\theta-\cos\theta+2+i(\sin3\theta-\sin\theta)$

$$\begin{aligned}|\,z^3-z+2\,|^2&=(\cos3\theta-\cos\theta+2)^2+(\sin3\theta-\sin\theta)^2\\&=6-2\left(\cos3\theta\cos\theta+\sin3\theta\sin\theta\right)+4\cos3\theta-4\cos\theta\\&=6-2\cos2\theta+4\cos3\theta-4\cos\theta\\&=4\left(4\cos^3\theta-\cos^2\theta-4\cos\theta+2\right)\end{aligned}$$

求导易得当 $\cos\theta=\dfrac{2}{3}$时 , $\left(|z^3-z+2\,|^2\right)_{\min}=\dfrac{8}{27}$.

(法二)

$$\begin{aligned}&\left|z^3-z+2\right|^2=\left|z^3-z+2z\bar{z}\right|^2=\left|z\left(z^2-1+2\bar{z}\right)\right|^2=\left|z^2-1+2\bar{z}\right|^2\\&=\left(z^2-1+2\bar{z}\right)\left(\bar{z}^2-1+2z\right)=4-2(\mathrm{z}+\bar{z})-\left(z^2+\bar{z}^2\right)+2\left(z^3+\bar{z}^3\right)\\&=4-2(z+\bar{z})-\left[(z+\bar{z})^2-2\right]+2(z+\bar{z})\left[(z+\bar{z})^2-3\right]\\&\xlongequal{\text{let },z+\bar{z}=x\in[-2,2]}2x^3-x^2-8x+6(\text{用导数易求最值})\end{aligned}$$

5 解答题

Solution.

(a) 设三次方程 $x^3+ax^2+bx+c=0$ 的三个实根分别为 $x_1,x_2,x_3\,(x_1\leqslant x_2\leqslant x_3)$ 由韦达定理, 可得 $x_1+x_2+x_3=-a,x_1x_2+x_2x_3+x_3x_1=b$, 注意到

$$\frac{1}{2}[2(\frac{x_3-x_1}{2})^2+(x_3-x_1)^2]\le a^2-3b=\frac{1}{2}\left[(x_1-x_2)^2+(x_2-x_3)^2+(x_3-x_1)^2\right]\le[(x_3-x_1)^2]$$

$$\Rightarrow \sqrt{a^2-3b} \leqslant x_3 - x_1 \leqslant 2\sqrt{\frac{a^2-3b}{3}}$$

当且仅当 $x_1 = x_2$ 或 $x_2 = x_3$ 时取到左端"=", 当且仅当 $x_1 + x_3 = 2x_2$ 时取到右端"=".

综上所述: $x_3 - x_1$ 的取值范围是 $\left[\sqrt{a^2-3b} \leqslant w \leqslant 2\sqrt{\frac{1}{3}a^2 - b}\right]$.

(b) 易知 $x^3 + ax^2 + bx + c = (x-a)(x-b)(x-c)$, 由韦达定理得

$$\begin{cases} a+b+c = -a \\ ab+bc+ca = b \\ abc = -c \end{cases}$$

$$\Rightarrow c = 0 \text{ or } ab = -1$$

若 $c = 0$, $\begin{cases} b = -2a \\ ab = b \end{cases}$, $\Rightarrow \begin{cases} a = 0 \\ b = 0 \end{cases}$ or $\begin{cases} a = 1 \\ b = -2 \end{cases}$.

若 $ab = -1$ 时, $\begin{cases} c = -2a - b \\ -1 + (a+b)c = b \end{cases}$, 带入原方程, 得:

$$b^4 + b^3 - 2b^2 + 2 = 0$$

$$\Rightarrow b + 1 = 0 \text{ or } b^3 - 2b + 2 = 0$$

若 $b + 1 = 0, \Rightarrow \begin{cases} a = 1 \\ c = -1 \end{cases}$

若 $b^3 - 2b + 2 = 0$,

由一元三次方程的求根公式得 $b = \sqrt[3]{\sqrt{\frac{19}{27}} - 1} - \sqrt[3]{\sqrt{\frac{19}{27}} + 1} \Rightarrow \begin{cases} a = -\frac{1}{b} \\ c = \frac{2}{2} - b \end{cases}$

综上, (a, b, c) 的值为

$$(0,0,0), \quad (1,-2,0), \quad (1,-1,-1), \quad \left(-\frac{1}{k}, k, \frac{2}{k} - k\right),$$

其中

$$k = \sqrt[3]{\sqrt{\frac{19}{27}} - 1} - \sqrt[3]{\sqrt{\frac{19}{27}} + 1}.$$

CHAPTER 15

2014 年启明考试数学部分

1 填空题 (本题共 5 小题, 每小题 8 分, 共 40 分)

1. 设 $f(x)=\dfrac{x}{\sqrt{1+x^2}}$, 则 n 重复合函数 $f_n(x)=f(f(\cdots f(x)\cdots))=$__________.
2. 设多项式 $p(x)$ 满足 $p\left(x^2+1\right)=(p(x))^2+1$ 和 $p(0)=0$, 则 $p(x)=$__________.
3. 设 $S_n=\displaystyle\sum_{k=1}^{n}\frac{6^k}{\left(3^{k+1}-2^{k+1}\right)\left(3^k-2^k\right)}$, 则极限 $\lim\limits_{n\to\infty}S_n=$__________.
4. 对 $x>0$, 函数 $f(x)=\dfrac{\left(x+\dfrac{1}{x}\right)^6-\left(x^6+\dfrac{1}{x^6}\right)-2}{\left(x+\dfrac{1}{x}\right)^3+\left(x^3+\dfrac{1}{x^3}\right)}$ 的最小值为__________.
5. 假设 20 名学生中的每一名学生可从提供的六门课程中选学一门至六门, 也可以一门都不选. 试判断下列命题是否正确: 存在 5 名学生和两门课程, 使得这 5 名学生都选了这两门课, 或者都没选, 选填“正确”或“否” __________.

2 证明题 (本题共 14 分)

1. 若 a 为正整数而 $\sqrt{a}$ 不为整数, 证明: $\sqrt{a}$ 为无理数.
2. 试证: 除 $\{0,0,0\}$ 外, 没有其他整数 m,n,p 使得

$$m+n\sqrt{2}+p\sqrt{3}=0.$$

3 证明题 (本题共 16 分)

设 a,b,c 为三角形三边之长, $p=\dfrac{a+b+c}{2}$, r 为内切圆半径, 证明:

$$\frac{1}{(p-a)^2}+\frac{1}{(p-b)^2}+\frac{1}{(p-c)^2}\geqslant\frac{1}{r^2}$$

4 证明题 (本题共 12 分)

证明: 设 m 是任一正整数, 则 $a_m=\dfrac{1}{2}+\dfrac{1}{3}+\dfrac{1}{4}+\dfrac{1}{5}+\cdots+\dfrac{1}{2^m}$ 不是整数

5 解答题 (本题共 18 分)

下图是 2013 年恒大足球俱乐部策划的主场对阵首尔 FC 足球队的亚冠决赛海报，左边是恒大队，右边是首尔队。该海报的寓意是什么？要求简单推导以下两个数学式子的结果：

$$\sqrt{1+2\sqrt{1+3\sqrt{1+4\sqrt{1+\cdots}}}} \qquad \text{和} \qquad \mathrm{e}^{\pi \mathrm{i}}+1.$$

已知欧拉公式 $\mathrm{e}^{\pi \mathrm{i}}=\cos\alpha+\mathrm{i}\sin\alpha$.

CHAPTER 16

2014 年启明考试数学部分参考答案

1　填空题 (每小题 8 分，共 40 分)

1. **Solution**. $f_n(x)=\dfrac{x}{\sqrt{1+nx^2}}$.
对 n 作数学归纳法. 当 $n=1$ 时结论成立；若

$$f_n(x)=\frac{x}{\sqrt{1+nx^2}},$$

则

$$f_{n+1}(x)=f\big(f_n(x)\big)=\frac{\frac{x}{\sqrt{1+nx^2}}}{\sqrt{1+\frac{x^2}{1+nx^2}}}=\frac{x}{\sqrt{1+(n+1)x^2}}.$$

因此结论对任意正整数 n 成立.

2. **Solution**. $p(x)=x$.
由 $p(0)=0$ 和题设恒等式，依次得到

$$p(1)=1,\qquad p(2)=2,\qquad p(5)=5,\quad \ldots$$

更一般地，令 $t_0=0$，$t_{n+1}=t_n^2+1$，则由归纳法可得 $p(t_n)=t_n$. 数列 $\{t_n\}$ 严格递增，所以多项式 $p(x)-x$ 有无穷多个零点，只能恒为零. 因此 $p(x)=x$.

3. **Solution**. 2.
注意到

$$\frac{6^k}{(3^{k+1}-2^{k+1})(3^k-2^k)}=\frac{2^k}{3^k-2^k}-\frac{2^{k+1}}{3^{k+1}-2^{k+1}}.$$

因而

$$S_n=2-\frac{2^{n+1}}{3^{n+1}-2^{n+1}}\longrightarrow 2.$$

4. **Solution**. 6.
令 $t=x+x^{-1}\geq 2$. 由 $x^3+x^{-3}=t^3-3t$ 可得

$$f(x)=\frac{t^6-(t^3-3t)^2}{t^3+(t^3-3t)}=3t\geq 6.$$

当且仅当 $x=1$ 时取等号.

5. **Solution**. 否.
令 20 名学生分别选择六门课程的 20 种三元子集. 对任意两门课程，恰有 $\dbinom{4}{1}=4$ 名学生同时选择它们，也恰有 4 名学生同时不选择它们. 因而不存在题述的 5 名学生，这给出了反例.

2 证明题 (共 14 分)

1. **Proof**. 设正整数 a 不是完全平方数. 在 a 的素因数分解中，至少有一个素数 q 的指数为奇数.
   若 $\sqrt{a}=\dfrac{m}{n}$，其中 m,n 为互素正整数，则 $m^2=an^2$. 等式左侧素因数 q 的指数为偶数，而右侧该指数等于 a 中 q 的奇指数与 n^2 中偶指数之和，仍为奇数，矛盾. 因此 $\sqrt{a}$ 是无理数.

2. **Proof**. 假设 $m+n\sqrt{2}+p\sqrt{3}=0$.
   若 $np\neq 0$，移项并平方得
   $$m^2=2n^2+3p^2+2np\sqrt{6},$$
   从而 $\sqrt{6}$ 为有理数，矛盾. 若 $np=0$，原式退化为一个有理数与一个无理数的有理倍数之和为零，只能相应系数均为零. 因此 $m=n=p=0$，除此之外不存在其他整数解.

3 证明题 (共 16 分)

Proof. 记三角形面积为 S. 由 $S=pr$ 和 Heron 公式，
$$p^2r^2=p(p-a)(p-b)(p-c).$$
令
$$x=\frac{1}{p-a},\qquad y=\frac{1}{p-b},\qquad z=\frac{1}{p-c}.$$
因为 $p=(p-a)+(p-b)+(p-c)$，所以
$$\frac{1}{r^2}=pxyz=xy+yz+zx.$$
于是原不等式等价于
$$x^2+y^2+z^2\geq xy+yz+zx,$$
而这由
$$x^2+y^2+z^2-xy-yz-zx=\frac{1}{2}\left[(x-y)^2+(y-z)^2+(z-x)^2\right]\geq 0$$
立即得到.

4 证明题 (共 12 分)

Proof. 记
$$a_m=\sum_{k=2}^{2^m}\frac{1}{k},\qquad L=\mathrm{lcm}(2,3,\ldots,2^m).$$
在 $L/2,L/3,\ldots,L/2^m$ 中，只有 $L/2^m$ 是奇数，其余各项均为偶数. 因此
$$La_m=\sum_{k=2}^{2^m}\frac{L}{k}$$
是奇数. 但 L 是偶数；若 a_m 为整数，则 La_m 应为 L 的整数倍，必为偶数，矛盾. 故 a_m 不是整数.

5 解答题 (共 18 分)

Solution. 第一式的值为 3，第二式的值为 0.

利用恒等式

$$n = \sqrt{1+(n-1)(n+1)}$$

逐层代入，有

$$3 = \sqrt{1+2\sqrt{1+3\sqrt{1+4\sqrt{1+\cdots}}}}.$$

另一方面，由 Euler 公式

$$\mathrm{e}^{\pi\mathrm{i}} = \cos\pi + \mathrm{i}\sin\pi = -1,$$

故 $\mathrm{e}^{\pi\mathrm{i}} + 1 = 0$. 因此海报所表达的比分为 $3:0$.

Part II

启明考试综合部分

CHAPTER 1

2025 年启明考试综合部分

致谢： 这是近年来最完整的一份综合试题. 感谢以下 2025 级同学参与试题重建：@ 放过孩纸、@SV、@ 鳄鱼、@Vision、@Sai1each、@AAA 水晶月球顶级试吃、@KARAS、@suzuran、@ 喃语、@ 疯传箱子、@asumi，以及所有参与整理的转专业交流群同学.

没有他们的帮助就没有这套试卷了.

1　应用：折叠比例

近年来，市场上出现了多款双屏折叠手机，在过往单屏手机的基础上，或通过“打开”形成大屏以获得更好的观看体验，或通过“合拢”以缩减尺寸获得更好的携带性. 尽管如此，视频和图片等固定长宽比的内容可能无法同时利用两块屏幕；例如，为完整显示画面，在小屏上不留空隙的视频，在大屏上需要填充黑边，导致实际画面增加得不够明显.

类似问题在印刷行业已有成熟的解决方案. 例如，本次考试的试卷用纸为纸张尺寸国际标准下的 A3 页，对折后即为常用的 A4 页，两者的长宽比几乎一致，使得一张 A3 页文档可以在保持比例的前提下，刚好横向缩小打印在一张 A4 页上.

Question. A4 页纸的长宽比是多少？

2　应用：概率产品

某在线游戏厂商推出新装备，玩家需要通过开箱子的方式收集 n 款碎片兑换一件装备，即在游戏过程中不断获得宝箱，每打开一个宝箱会随机出一款碎片.

Question 1. 当 $n = 2$ 且一款碎片出现概率是另一款的两倍时，玩家平均需要开多少箱子才能集齐？

Question 2. 如果每款碎片出现的概率相同，玩家平均需要开多少箱子才能集齐全部 n 款？

3　折扣定价与分批购买

商家在线销售某产品，为促进销量，将其定为：每件 5 元，每两件 8 元，以及每三件 p 元（为简便计，仅假定 p 为自然数）. 如果消费者购买三件以上，可以分批多次购买. 例如需要四件时，既可以分 4 批购买每次购买一件，也可以分 2 批每次购买两件，还可以先买一件再购买三件. 显然，不同的分批方案对应不同的总成本，而聪明的消费者会采用总成本最低的分批方案.

请结合上述背景，回答如下问题：

Question 1. 商家在确定折扣价格时，会设定 $p \geq 9$，否则只需两件的消费者最终会购买三件. 类似地，商家还会设定 $p \leq 12$，这是为什么？（10 分）

Question 2. 消费者打算购买 n 件时的最优分批方案是什么？请先建立适当的数学模型，分析并计算相应的最低总成本函数 $C(n)$，并用简单文字描述最优分批方案.（15 分）

Question 3. 为提高消费者满意度，商家可以进一步允许消费者退货. 但如果设定了较高的退货价格 r，比如 $r > 5$，会给消费者套利的机会，即通过买入 1 件后再退货得到 $r - 5$ 的收益. 为避免出现这种情况，退货价格不能超过多少？（5 分）

4 第四题：定比投资与 Kelly Criterion

某人采用定比方式投资到某类理财产品，即每期期初投入固定比例为 x 的持有资金. 假设各期结束后有概率 p 会产生收益 $xW > x$，有概率 q 产生损失 $xL < x$，有概率 $1 - p - q$ 不赚不亏. 更新持有资金后，下期仍滚动投资同样比例的资金.

该投资者希望在长期获得较好回报率和相应风险. 如果每次都投入全部资金（即 $x = 1$），可能某次损失就会导致大量亏损；如果是投资过于保守（即 x 很小），投资回报增长又会太慢. 因此，有必要设计合理的投资比例. 假设初始资金为 1，并滚动式投资 n 期，请回答如下问题：

Question 1. 当 n 很大时，平均有多少期产生收益，多少期产生损失？（10 分）

Question 2. 第 n 期结束时，平均持有资金可记为投资比例 x 的函数 $V(x)$，请给出 $V(x)$ 的表达式，求解最优投资比例 x^*，并用文字简单描述 p, q, L, W 的变化对 x^* 的影响.（15 分）

Question 3. 投资者也可采用定额方式投资. 当其初始资金为 m 份时，每期可选择投入 1 份或 0 份，直到第 n 期结束或持有资金归零为止. 在上述设定下，帮助其决策每期投入的资金.

CHAPTER 2

2025 年启明考试综合部分参考答案

1 第一题

Solution. $\sqrt{2}$. 设同系列纸张的长: 短比为 $r : 1$ $(r > 1)$. 沿与短边平行的中线对折，长边被一分为二，短边不变；对折后得到一张尺寸为 $\frac{r}{2} \times 1$ 的纸. 其长边为 $\max\{1, \frac{r}{2}\}$，短边为 $\min\{1, \frac{r}{2}\}$. 当 $r < 2$（真实纸张定制显然 $r < 2$）时，新纸的长宽比为

$$\frac{长}{短} = \frac{1}{r/2} = \frac{2}{r}.$$

若对折后与原纸相似，则两者的长宽比相等：$r = \frac{2}{r}$. 解得

$$r^2 = 2 \quad \Rightarrow \quad \boxed{r = \sqrt{2}}.$$

2 第二题

Question 1

Solution. $\frac{7}{2}$. 设两种碎片分别为 A 与 B，且 $\mathbb{P}(B) = 2/3$, $\mathbb{P}(A) = 1/3$. 第一次开箱必得新类型（期望 1 次）. 之后需要等待第一次未见过的那一类出现，若先抽到 B（概率 $2/3$），则等待 A 的几何分布期望为 $1/\mathbb{P}(A) = 3$；若先抽到 A（概率 $1/3$），则等待 B 的期望为 $1/\mathbb{P}(B) = 3/2$. 于是总期望为

$$\mathbb{E}[T] = 1 + \frac{2}{3} \cdot 3 + \frac{1}{3} \cdot \frac{3}{2} = 1 + 2 + \frac{1}{2} = \boxed{\frac{7}{2}}.$$

Question 2

Solution. nH_n. 经典“集齐优惠券”模型. 记收齐前所需开箱数为 T_n. 第 k 种新碎片（已收集 $k-1$ 种后）出现的成功概率为 $(n-(k-1))/n$，对应的几何分布期望为 $\frac{n}{n-k+1}$，线性叠加得

$$\mathbb{E}[T_n] = \sum_{k=1}^{n} \frac{n}{k} = n\left(1 + \frac{1}{2} + \cdots + \frac{1}{n}\right) = \boxed{nH_n},$$

其中 H_n 为第 n 个调和数. 大样本近似：$\mathbb{E}[T_n] \approx n(\ln n + \gamma) + \frac{1}{2} + o(1)$.

3 第三题：折扣定价与分批购买

单价 5 元，二件装 8 元，三件装 p 元（$p \in \mathbb{N}$）. 可分批购买以恰好买到 n 件.

Question 1

Solution. $9 \le p \le 12$.

- 若 $p \le 8$，则对仅需两件的消费者来说，“三件装 p 元”不比“两件装 8 元”贵（甚至更便宜），他们会转而购买三件，从而“只需两件最终买三件”. 为避免此侵蚀，商家会设 $p \ge 9$.
- 若 $p \ge 13$，买三件用“二件装 + 单件”花费 $8+5=13$ 元不更贵（或更便宜），三件装缺乏吸引力. 为体现实质优惠（严格优于 13 元），商家会设 $p \le 12$.

故合理区间为 $\boxed{9 \le p \le 12}$.

Question 2

Solution. 建立整数规划模型如下.

精确定义的优化模型. 设购买 a 个单件、b 个二件装、c 个三件装，满足

$$a + 2b + 3c = n, \quad a, b, c \in \mathbb{Z}_{\ge 0},$$

总成本为 $5a + 8b + pc$. 最低成本函数

$$\boxed{C(n) = \min_{c=0,\dots,\lfloor n/3 \rfloor} \left\{ pc + 8\left\lfloor \frac{n-3c}{2} \right\rfloor + 5\big((n-3c) \bmod 2\big) \right\}}.$$

该式总是正确（任意 n、任意 p），且可直接据此计算与实现.

在 $9 \le p \le 12$ 的显式解（$n \ge 3$）. 此时三件装的单位价 $p/3 \le 4$，不劣于二件装. 令 $n = 3k, r = n \bmod 3$，则

$$C(n) = \begin{cases} kp, & r = 0; \\ kp + \min\{5,\ 16 - p\}, & r = 1; \\ kp + 8, & r = 2. \end{cases}$$

并且最优分批方案为:

- $r = 0$: 只买三件装（k 个）.
- $r = 1$: 若 $p \le 11$，买 k 个三件装再补 1 个单件；若 $p = 12$，用 $(k-1)$ 个三件装 + 两个二件装（避免买单件）.
- $r = 2$: 买 k 个三件装 + 一个二件装（当 $p = 12$ 时，亦可用 $(k-2)$ 个三件装 + 四个二件装，成本相同）.

边界小样本修正： $C(1) = 5,\ C(2) = 8$.

Question 3

Solution. $r \le \min\left\{4, \dfrac{p}{3}\right\}$. 若允许按件退货且退货价为 r，消费者可“买入套餐再逐件退”. 为杜绝无风险套利，需保证对每一种可买入的按件成本 $\bar{c}$ 都有 $r \le \bar{c}$. 三种来源的按件成本分别为

$$5, \quad \frac{8}{2} = 4, \quad \frac{p}{3}.$$

故无套利条件

$$\boxed{r \ \le\ \min\left\{5,\ 4,\ \frac{p}{3}\right\} \ =\ \min\left\{4,\ \frac{p}{3}\right\}}.$$

在 $9 \le p \le 12$ 时即 $r \le p/3\ (\le 4)$.

4 第四题：定比投资与 Kelly 准则

每期将资金的固定比例 $x \in [0,1]$ 投入风险资产；期末随机结果为

$$G(x) = \begin{cases} 1 + x(W-1), & \text{概率} p, \\ 1 + x(L-1), & \text{概率} q, \\ 1, & \text{概率} 1-p-q, \end{cases} \qquad (W > 1,\ 0 \le L < 1).$$

初始资金 $V_0 = 1$，独立同分布.

Question 1

Solution. 获利、亏损和不涨不跌的期数分别约为 pn、qn 和 $(1-p-q)n$. 当 n 很大时，

$$\boxed{\text{获利期数} \approx pn,\ \text{亏损期数} \approx qn,\ \text{不涨不跌期数} \approx (1-p-q)n}.$$

Question 2

Solution. 分别从期望资金和长期几何增长率两个角度讨论.

（一）"平均资金"（期望）视角. 由于各期独立，

$$\mathbb{E}[V_n(x)] = \big(\mathbb{E}[G(x)]\big)^n = \Big(1 + x\big[p(W-1) + q(L-1)\big]\Big)^n.$$

若 $p(W-1) + q(L-1) > 0$，则期望单调随 x 增大而增，最优为 $x = 1$；若 < 0，最优为 $x = 0$；若 $= 0$，任意 x 等价. 该结论虽简单，但并不控制长期波动风险.

（二）Kelly 准则（最大化长期几何增长率）. 定义长期平均对数增长率

$$g(x) = \mathbb{E}[\ln G(x)] = p\ln\big(1 + x(W-1)\big) + q\ln\big(1 + x(L-1)\big),$$

（第三种结果取 $\ln 1 = 0$）. 一阶条件

$$g'(x) = \frac{p(W-1)}{1 + x(W-1)} + \frac{q(L-1)}{1 + x(L-1)} = 0.$$

令 $a = W - 1 > 0$, $b = 1 - L > 0$（则 $L - 1 = -b$），解得

$$\boxed{x^*_{\text{Kelly}} = \frac{p(W-1) - q(1-L)}{(W-1)(1-L)\,(p+q)}}.$$

可见 x^* 随 p、W 增大而升高，随 q 增大或 L 降低（亏损更深，即 $1-L$ 增大）而降低. 实际应用中需限制 x 在可行域

$$0 \le x < \frac{1}{1-L} \quad (\text{保证} 1 + x(L-1) > 0),$$

并常剪裁到 $[0,1]$：$x^* = \max\{0, \min\{1, x^*_{\text{Kelly}}\}\}$.

Question 3

Solution. 以最大化期望终值为目标，可使用如下动态决策. 初始资金为 m（离散 "份"），每期可投入 0 或 1 份. 设目标为最大化期望终值. 记期 t 起始资金为 S，若本期投入 1 份，则期末资金的期望为

$$\mathbb{E}[S'] = S - 1 + \big[pW + qL + (1-p-q)\cdot 1\big] = S + \underbrace{\big[p(W-1) + q(L-1)\big]}_{\mu}.$$

由动态规划的线性性可归纳得最优策略为

$$\boxed{\text{若} \mu := p(W-1) + q(L-1) > 0,\ \text{且} S \ge 1,\ \text{则当期投入} 1\ \text{份；若} \mu \le 0,\ \text{则不投.}}$$

CHAPTER 3

2024 年启明考试综合部分

1 第一题

我们考虑一个生产商和零售商的定价问题. 生产成本为 c，最高售价为 a，销量为 $a-p$（p 为实际售价）.

合作情况（π_0）：生产商和零售商合作，共同决定零售价 p，

$$\pi_0 = \max_{p\in[c,a]} (p-c)(a-p).$$

非合作情况（π_1）：生产商先确定批发价 $\widetilde{w}$，零售商再确定零售价 p，

$$\pi^r(\widetilde{w}) = \max_{p\in[\widetilde{w},a]} (p-\widetilde{w})(a-p),$$

$$\pi^s = \max_{\widetilde{w}\in[c,a]} (\widetilde{w}-c)(a-p_{\widetilde{w}}), \qquad \pi_1 = \pi^r(\widetilde{w}) + \pi^s.$$

其中，$\pi^r(\widetilde{w})$ 是零售商的最大利润，π^s 是生产商的最大利润.

问题目标：求 $\rho = \dfrac{\pi_1}{\pi_0}$ 的值.

2 第二题

仓储系统中有 n 件不同物品. 货位参数为 $a_i\,(a_1 < a_2 < \cdots < a_n)$, 出货频次为 $x_i\,(x_1 < x_2 < \cdots < x_n)$. 对一个排列 σ，总出货时间可表示为

$$T(\sigma) = \sum_{i=1}^{n} a_i x_{\sigma(i)}.$$

1. 将上述实际问题转化为最优化问题.
2. 证明：出货频率越高的货物放在越前面，才能使总出货时间最短（提示：逐步交换）.
3. 证明：出货频率越高的货物放在越后面，才能使总出货时间最大（提示同上）.

3 第三题

工业生产中经常需要按批次制备原料，将其维持在特定状态，再按特定速率连续加入制备反应釜中，假设每制备一批原料的成本为 k (与数量无关), 将单位数量原料维持单位时间的成本为 h 原料从 0 时刻开始到 t 时刻的累积消耗量为 $D_t = t^{\frac{1}{n}}$. 其中累积生产时长 $t \le 1$，参数 $n \ge 1$，表明原料消耗速率随反应时长不变或递减，(例如 $n=1$ 时 $Dt=t$, 匀速消耗原料).

Question 1. 给出函数 $y = D_t$ 在 $t \in [0,1]$ 上的示意图，合理展示该曲线的趋势特点.

在保证持续生产和避免浪费的前提下，为了尽可能的减少成本，有必要选择合理的原料制备方案，如果整个过程中只制备一次，那么显然应在 0 时刻制备 D_1 数量的原料，令其逐渐消耗至 $t=1$ 时刻，因此单次制备方案的成本为 $k+hI$，其中 I 表示处于线段 $y=D_1$ 和曲线 $y=D_t$ 之间的面积.

Question 2. 如果在 0 和 t 时刻各制备一次，两次应分别制备多少原料，才能使总成本最小?

当批次制备时刻和批量都可以灵活调节时，直观上如果 k 相对 h 较小，应采用小批量，多批次的方式，而当 k 相较于 h 较大时，应利用大批量少批次的方式. 甚至只制备一次.

Question 3. 已知当且仅当 $k<A_n h$ 时，两批次方案严格优于单批次方案. 请给出 A_n 的具体表达式.

提示： 函数 $x\left(c-x^{1/n}\right)$ 在 $x\in[0,c^n]$ 上的最大值为 $\dfrac{(cn)^n}{(n+1)^{n+1}}$.

Question 4. 当 $n=1$ 和 $n=2$ 时，两批次方案严格优于三批次方案的充要条件是什么?

CHAPTER 4

2023 年启明考试综合部分

1 优化与建模

1. 封闭曲线 A 中有一条长为 1 的弦，弦上一点 P 将其分为两段，长度分别为 x 和 $1-x$. 该弦在曲线上转过一周，P 随弦同时转动，形成一条封闭曲线 B. 曲线 A 与曲线 B 围成的面积之差为 $S(x)$. 由 Pappus–Guldin 原理可知，$S(x)$ 与曲线 A, B 的形状无关，仅与点 P 的位置有关. 求 $S(x)$ 的最大值及此时 x 的取值.（各 10 分）

2. 某商店的称量天平损坏，其左臂比右臂长 10% ∼ 20%. 一位顾客购买 2 kg 物品，商家分别以“左物右码”和“右物左码”的方式称量 1 kg 物品. 最终是商家还是顾客遭受损失？损失比例的范围是什么？（各 10 分）

2 信息熵

信息熵（information entropy）是信息论的基本概念，用于描述信息源各可能事件发生的不确定性.20 世纪 40 年代，Shannon 借鉴热力学中的概念，把信息中排除冗余后的平均信息量称为“信息熵”，并给出如下定义：

$$H(x)=\sum_{i=1}^{n}-p\left(x_i\right)\log_a p(x_i)$$

$n=2$ 时，上式可写为

$$H(p,q)=-p\log_a p-q\log_a q$$

当 $p=q=0.5$ 时，上式取得极值. 回答下列问题：

1. 比特（bit, binary digit）是信息量单位. 若抛硬币时正面与反面朝上的概率均为 $\dfrac{1}{2}$，根据公式算得信息量为 1 bit，此时底数 a 应取何值？并给出合理解释.

2. 有人提出，对于相互独立的两个随机变量 X, Y，应有 $H(X+Y)=H(X)+H(Y)$. 设 X, Y 均服从参数为 p 的两点分布，则 $Z=X+Y$ 服从参数为 $2, p$ 的二项分布. 判断 $H(Z)=2H(X)$ 是否成立，并说明理由.

3. 利用 Jensen 不等式证明 $H(X)$ 存在最大值.
   提示：若 $f(x)$ 为凸函数，则

$$f(a_1x_1+\cdots+a_nx_n)\leq a_1f(x_1)+\cdots+a_nf(x_n),\qquad \sum_{k=1}^{n}a_k=1.$$

4. 现有 2023 个小球（编号为 $1,2,\ldots,2023$）和一个足够大的天平，其中一个劣质小球略轻. 至少需要多少次称量才能找出该球？

3 产量分配

某公司拟使用 t 万元采购 n 种原料，各种原料的单价和采购量分别记为

$$(p_1, p_2, \ldots, p_n), \qquad (x_1, x_2, \ldots, x_n).$$

产品总产量为

$$\min\{q_1 x_1, q_2 x_2, \ldots, q_n x_n\}.$$

在尽量扩大产量的前提下,该公司应采用怎样的采购策略?若预算翻倍,哪一种原料增加的采购量最多?(各 10 分)

Part III

启明考试英语部分

CHAPTER 1

2025 年启明考试英语部分

Acknowledgement: Many thanks to these students in the Fall 2025 cohort for their outstanding contributions:

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Part I. Listening

Listening 1: TOEFL 67

Listening 2: TOEFL 67

Listening 3: TOEFL 67

Listening 4: TOEFL 62

Listening 5: TOEFL 53

Part II. Reading Comprehension

Passage 1

Source: 2020 CET-6 Sept.(Version 2) Sec.C Passage 1

Organic agriculture is a relatively untapped resource for feeding the Earth's population, especially in the face of climate change and other global challenges. That's the conclusion I reached in reviewing 40 years of science comparing the long-term prospects of organic and conventional farming.

The review study, "Organic Agriculture in the 21st Century," is featured as the cover story for the February issue of the journal Nature Plants. It is the first to compare organic and conventional agriculture across the main goals of sustainability identified by the National Academy of Sciences: productivity, economics and environment. Critics have long argued that organic agriculture is inefficient, requiring more land to yield the same amount of food. It's true that organic farming produces lower yields, averaging 10 to 20 percent less than conventional. Advocates contend that the environmental advantages of organic agriculture far outweigh the lower yields, and that increasing research and breeding resources for organic systems would reduce the yield gap. Sometimes excluded from these arguments is the fact that we already produce enough food to more than feed the world's 7.4 billion people but do not provide adequate access to all individuals.

In some cases, organic yields can be higher than conventional. For example, in severe drought conditions, which are expected to increase with climate change in many areas, organic farms can produce as good, if not better, yields because of the higher water-holding capacity of organically farmed soils. What science does tell us is that mainstream conventional farming systems have provided growing supplies of food and other products but often at the expense of other sustainability goals.

Conventional agriculture may produce more food, but it often comes at a cost to the environment. Biodiversity loss, environmental degradation, and severe impacts on ecosystem services have not only accompanied conventional farming systems but have often extended well beyond their field boundaries. With organic agriculture, environmental costs tend to be lower and the benefits greater. Overall, organic farms tend to store more soil carbon, have better soil quality and reduce soil erosion compared to their conventional counterparts. Organic agriculture also creates less soil and water pollution and lower greenhouse gas emissions. And it's more energy-efficient because it doesn't rely on synthetic fertilizers or pesticides.

Organic agriculture is also associated with greater biodiversity of plants, animals, insects and microorganisms as well as genetic diversity. Biodiversity increases the services that nature provides and improves the ability of farming systems to adapt to changing conditions.

Despite lower yields, organic agriculture is more profitable for farmers because consumers are willing to pay more. Higher prices, called price premiums, can be justified as a way to compensate farmers for providing ecosystem services and avoiding environmental damage or external costs.

1. What do we learn from the conclusion of the author's review study?

A. More resources should be tapped for feeding the world's population.

B. Organic farming may be exploited to solve the global food problem.

C. The long-term prospects of organic farming are yet to be explored.

D. Organic farming is at least as promising as conventional farming.

2. What is the critics' argument against organic farming?

A. It cannot meet the need for food. B. It cannot increase farm yields.

C. It is not really practical. D. It is not that productive.

3. What does the author think should be taken into account in arguing about organic farming?

A. Growth in world population. B. Deterioration in soil fertility.

C. Inequality in food distribution. D. Advance in farming technology.

4. What does science tell us about conventional farming?

A. It will not be able to meet global food demand.

B. It is not conducive to sustainable development.

C. It will eventually give way to organic farming.

D. It is going mainstream throughout the world.

5. Why does the author think higher prices of organic farm produce are justifiable?

A. They give farmers going organic a big competitive edge.

B. They motivate farmers to upgrade farming technology.

C. Organic farming costs more than conventional farming.

D. Organic farming does long-term good to the ecosystem.

Passage 2

Source: 2020 CET-6 Sept.(Version 1) Sec.C Passage 2

Home to virgin reefs, rare sharks and vast numbers of exotic fish, the Coral Sea is a unique haven of biodiversity off the northeastern coast of Australia. If a proposal by the Australian government goes ahead, the region will also become the world’s largest marine protected area, with restrictions or bans on fishing, mining and marine farming. The Coral Sea reserve would cover almost 990,000 square kilometres and stretch as far as 1,100 kilometres from the coast. Unveiled recently by environment minister Tony Burke, the proposal would be the last in a series of proposed marine reserves around Australia’s coast.

But the scheme is attracting criticism from scientists and conservation groups, who argue that the government hasn’t gone far enough in protecting the Coral Sea, or in other marine reserves in the coastal network. Hugh Possingham, director of the Centre of Excellence for Environmental Decisions at the University of Queens-land, points out that little more than half of the Coral Sea reserve is proposed as “no take” area, in which all fishing would be banned. The world’s largest existing marine reserve, established last year by the British government in the Indian Ocean, spans 554,000 km2 and is a no-take zone throughout. An alliance of campaigning conservation groups argues that more of the Coral Sea should receive this level of protection.

“I would like to have seen more protection for coral reefs,” says Terry Hughes, director of the Centre of Excellence for Coral Reef Studies at James Cook University in Queensland. “More than 20 of them would be outside the no-take area and vulnerable to catch-and-release fishing”.

As Nature went to press, the Australian government had not responded to specific criticisms of the plan. But Robin Beaman, a marine geologist at James Cook University, says that the reserve does “broadly protect the range of habitats” in the sea. “I can testify to the huge effort that government agencies and other organisations have put into trying to understand the ecological values of this vast area,” he says.

Reserves proposed earlier this year for Australia’s southwestern and northwestern coastal regions have also been criticised for failing to give habitats adequate protection. In August, 173 marine scientists signed an open letter to the government saying they were “greatly concerned” that the proposals for the southwestern region had not been based on the “core science principles” of reserves—the protected regions were not, for instance, representative of all the habitats in the region, they said.

Critics say that the southwestern reserve offers the greatest protection to the offshore areas where commercial opportunities are fewest and where there is little threat to the environment, a contention also levelled at the Coral Sea plan.

1. What do we learn from the passage about the Coral Sea?

A. It is exceptionally rich in marine life.
B. It is the biggest marine protected area.
C. It remains largely undisturbed by humans.
D. It is a unique haven of endangered species.

2. What does the Australian government plan to do according to Tony Burke?

A. Make a new proposal to protect the Coral Sea.
B. Revise its conservation plan owing to criticisms.
C. Upgrade the established reserves to protect marine life.
D. Complete the series of marine reserves around its coast.

3. What is scientists’ argument about the Coral Sea proposal?

A. The government has not done enough for marine protection.
B. It will not improve the marine reserves along Australia’s coast.
C. The government has not consulted them in drawing up the proposal.
D. It is not based on sufficient investigations into the ecological system.

4. What does marine geologist Robin Beaman say about the Coral Sea plan?

A. It can compare with the British government's effort in the Indian Ocean.

B. It will result in the establishment of the world's largest marine reserve.

C. It will ensure the sustainability of the fishing industry around the coast.

D. It is a tremendous joint effort to protect the range of marine habitats.

5. What do critics think of the Coral Sea plan?

A. It will do more harm than good to the environment.

B. It will adversely affect Australia's fishing industry.

C. It will protect regions that actually require little protection.

D. It will win little support from environmental organisations.

Passage 3

Source: TEST FOR ENGLISH MAJOR (2023): GRADE FOUR

(1) Pairing the words "baby" and "sleep" can evoke strong emotions. Those who have had limited contact with little ones might interpret this word–combination as implying deep and prolonged slumber. For others, this union of words may elicit memories of prolonged periods of chaotic sleep (or what can feel like no sleep at all).

(2) Coping with the way babies sleep can be difficult. It's not that babies don't sleep. In fact, they sleep more than at any other stage of life. It's more an issue of when they sleep. Newborns start by sleeping and waking around the clock. This is not always easy for parents. There is even research suggesting that, in adults, waking repeatedly at night can feel as bad as getting hardly any sleep in terms of attentional skills, fatigue levels and symptoms of depression.

(3) As to why infants wake at night, this is best explained by thinking about the two things that govern our sleep: the homeostatic and circadian processes. The crux of the homeostatic process is the straightforward idea that the longer we have been awake the greater our sleep drive (and the more sleepy we feel). It may take an adult an entire day to build up enough sleep drive to fall asleep at bedtime, but an infant may only need an hour or two of wakefulness before being able to drift off to sleep.

(4) The second process is circadian, which works like a clock. Adults typically feel more awake during the morning hours and sleepy at night, regardless of when we last slept. In very young babies this process is not yet developed. This means that sleep is more likely to occur at different points across the 24-hour day.

(5) Practically speaking, the immaturity of these two processes means that newborn babies are actually expected to wake at night: they are doing exactly what they are supposed to do! They start life with small stomachs which need to be filled regularly so your child can gain strength and stay hydrated, so it's a good thing that they are waking regularly to feed. As hard as disturbed sleep can be for caregivers, a waking baby is a good thing.

(6) But how long might this continue? The parenting mantra "this too shall pass" is true when it comes to dealing with certain aspects of a baby's sleep. Night wakings typically become less frequent as an infant grows up, and sleep changes in other ways throughout a person's life. For example, sleep length reduces and there are changes in sleep architecture (or composition of Rapid Eye Movement, REM, and Non-Rapid Eye Movement, NREM, sleep). A premature baby's sleep cycle might take just 45 minutes, whereas an adult's can be double that at 90 minutes. Other changes also occur; for example, whereas babies' sleep cycles start with REM-like sleep, adults start with NREM sleep.

(7) For some parents, knowing that sleep changes throughout life is enough to help them cope with an infant's night awakenings. In fact, sleep education alone can help some parents to deal with infant sleep. Other parents want more detailed information, such as about babies' sleep schedules, bedtime routines, sleep problems, sleep safety, sleep environment, naps, sleep training, and special circumstances. We provide all this information, and

more, on our website, where physicians, psychologists, and researchers from the Pediatric Sleep Council have also answered hundreds of questions about sleep via video and text.

(8) Remember: sleep matters. Experts agree that sleep is essential for health, growth and general development. It is important not just for babies, but for parents too. If we can improve sleep within a family, and create a happy association between the words "baby" and "sleep", then the Pediatric Sleep Council has achieved its goal.

41. According to the passage, the difficulty in handling baby sleep lies in not knowing

A. when they switch to slumber .
B. how long they sleep well
C. how they fall asleep
D. when they sleep

42. In explaining the circadian process, the author has used more than other means.

A. simile
B. comparison
C. metaphor
D. exaggeration

43. Which of the following statements is CORRECT about baby sleep?

A. Parents need to be well informed about baby sleep.
B. Babies' sleep cycle tends to be similar to that of adults.
C. A happy link between "baby" and "sleep" is impossible.
D. Babies sleep a lot more during daytime than nighttime.

44. Which of the following is the most appropriate title for the passage?

A. Secrets of parenting babies.
B. Baby sleep patterns.
C. Does baby sleep matter?
D. What are the must-knows of sleep?

Passage 4: Evidence of the Earliest Writing

Although literacy appeared independently in several parts of the prehistoric world, the earliest evidence of writing is the cuneiform Sumerian script on the clay tablets of ancient Mesopotamia, which, archaeological detective work has revealed, had its origins in the accounting practices of commercial activity. Researchers demonstrated that preliterate people, to keep track of the goods they produced and exchanged, created a system of accounting using clay tokens as symbolic representations of their products. Over many thousands of years, the symbols evolved through several stages of abstraction until they became wedge-shaped (cuneiform) signs on clay tablets, recognizable as writing.

The original tokens (circa 8500 B.C.E.) were three-dimensional solid shapes—tiny spheres, cones, disks, and cylinders. A debt of six units of grain and eight head of livestock, for example, might have been represented by six conical and eight cylindrical tokens. To keep batches of tokens together, an innovation was introduced (circa 3250 B.C.E.) whereby they were sealed inside clay envelopes that could be broken open and counted when it came time for a debt to be repaid. But because the contents of the envelopes could easily be forgotten, two-dimensional representations of the three-dimensional tokens were impressed into the surface of the envelopes before they were sealed. Eventually, having two sets of equivalent symbols—the internal tokens and external markings—came to seem redundant, so the tokens were eliminated (circa 3250–3100 B.C.E.), and only solid clay tablets with two-dimensional symbols were retained. Over time, the symbols became more numerous, varied, and abstract and came to represent more than trade commodities, evolving eventually into cuneiform writing.

The evolution of the symbolism is reflected in the archaeological record first of all by the increasing complexity of the tokens themselves. The earliest tokens, dating from about 10,000 to 6,000 years ago, were of only the simplest geometric shapes. But about 3500 B.C.E., more complex tokens came into common usage, including many naturalistic forms shaped like miniature tools, furniture, fruit, and humans. The earlier, plain tokens were counters for agricultural products, whereas the complex ones stood for finished products, such as bread, oil, perfume, wool, and rope, and for items produced in workshops, such as metal, bracelets, types of cloth, garments, mats, pieces of furniture, tools, and a variety of stone and pottery vessels. The signs marked on clay tablets likewise evolved from simple wedges, circles, ovals, and triangles based on the plain tokens to pictographs derived from the complex tokens.

Before this evidence came to light, the inventors of writing were assumed by researchers to have been an intellectual elite. Some, for example, hypothesized that writing emerged when members of the priestly caste agreed among themselves on written signs. But the association of the plain tokens with the first farmers and of the complex tokens with the first artisans—and the fact that the token-and-envelope accounting system invariably represented only small-scale transactions—testifies to the relatively modest social status of the creators of writing.

And not only of literacy, but numeracy (the representation of quantitative concepts) as well. The evidence of the tokens provides further confirmation that mathematics originated in people's desire to keep records of flocks and other goods. Another immensely significant step occurred around 3100 B.C.E., when Sumerian accountants extended the token-based signs to include the first real numerals. Previously, units of grain had been represented by direct one-to-one correspondence—by repeating the token or symbol for a unit of grain the required number of times. The accountants, however, devised numeral signs distinct from commodity signs, so that eighteen units of grain could be indicated by preceding a single grain symbol with a symbol denoting "18." Their invention of abstract numerals and abstract counting was one of the most revolutionary advances in the history of mathematics.

What was the social status of the anonymous accountants who produced this breakthrough? The immense volume of clay tablets unearthed in the ruins of the Sumerian temples where the accounts were kept suggests a social differentiation within the scribal class, with a virtual army of lower-ranking tabulators performing the monotonous job of tallying commodities. We can only speculate as to how high or low the inventors of true numerals were in the scribal hierarchy, but it stands to reason that this laborsaving innovation would have been the brainchild of the lower-ranking types whose drudgery it eased.

Part III. Cloze Test（完形填空）

The shorter growing seasons expected with climate change over the next 40 years will endanger hundreds of millions of already poor people in the global tropics, say researchers working __62__ the world's leading agricultural organisations.

The effects of climate change are likely to be seen across the entire tropical __63__, but many areas previously considered to be __64__ food secure are likely to become highly __65__ to droughts, extreme weather and higher temperatures, say the __66__ with the Consultative Group on International Agricultural Research.

Intensively farmed areas __67__ northeast Brazil and Mexico are likely to see their __68__ growing seasons fall below 120 days, which is __69__ for crops such as corn to mature. Many other places in Latin America are likely to __70__ temperatures that are too hot for bean __71__, a staple in the region.

The impact could be __72__ most in India and southeast Asia. More than 300 million people in south Asia are likely to be affected even with a 5% decrease in the __73__ of the growing season.

Higher peak temperatures are also expected to take a heavy __74__ on food producers. Today there are 56 million crop-dependent people in parts of west Africa and India who live in areas where, in 40 years, maximum daily temperatures could be higher than 30°C. This is __75__ to the maximum temperature that beans can tolerate, __76__ corn and rice yields suffer when temperatures __77__ this level.

"We are starting to see much more clearly __78__ the effects of climate change on agriculture could __79__ hunger and poverty," said research leader Patti Kristjanson. "Farmers already adapt __80__ variable weather by changing their planting schedules. What this study suggests is that the speed of climate __81__ and the magnitude of the changes required to adapt could be much greater."

62. A. with B. from C. at D. for

63. A. zone B. belt C. region D. area

64. A. relatively B. perfectly C. completely D. marginally

65. A. vulnerable B. resistant C. indifferent D. immune

66. A. researchers B. officials C. farmers D. journalists

67. A. like B. as C. to D. for

68. A. prime B. average C. minor D. annual

69. A. critical B. optional C. excessive D. minimal

70. A. experience B. expose C. harvest D. forecast

71. A. production B. planting C. cultivation D. storage

72. A. felt B. seen C. noticed D. suffered

73. A. length B. level C. scale D. volume

74. A. toll B. role C. load D. burden

75. A. close B. equal C. similar D. contrary

76. A. while B. if C. unless D. because

77. A. exceed B. approach C. reach D. fall below

78. A. where B. how C. that D. whether

79. A. intensify B. weaken C. relieve D. eliminate

80. A. to B. in C. with D. for

81. A. shifts B. change C. fluctuations D. impacts

CHAPTER 2

2024 年启明考试英语部分

Notice: Most of the questions in this document were obtained by seniors after copying the original text and searching the original text outside the examination room. The questions are different from the examination questions and are for reference only.

Part I. Listening

Lack of materials. This part is similar to CET-6 Listening part.

Part II. Reading Comprehension

Passage One

Social media can be a powerful communication tool for employees, helping them to collaborate, share ideas and solve problems. Research has shown that 82% of employees think social media can improve work relationships and 60% believe it can support decision-making processes. These beliefs contribute to a majority of workers connecting with colleagues on social media, even during work hours.

Employers typically worry that social media is a productivity killer; more than half of U.S. employers reportedly block access to social media at work. In my research with 277 employees of a healthcare organization I found these concerns to be misguided. Social media doesn't reduce productivity nearly as much as it kills employee retention.

In the first part of the study I surveyed the employees about why and how they used platforms like Facebook, Twitter, or LinkedIn. Respondents were then asked about their work behaviors, including whether they felt motivated in their jobs and showed initiative at work. I found employees who engage in online social interactions with co-workers through social media blogs tend to be more motivated and come up with innovative ideas. But when employees interact with individuals outside the organization, they are less motivated and show less initiative. In the second part of the study I found 76% of employees using social media for work took an interest in other organizations they found on social media. When I examined how respondents expressed openness to new careers and employers, I found that they engaged in some key activities including researching new organizations and making new work connections. These findings present a dilemma for managers: employees using social media at work are more engaged and more productive, but they are also more likely to leave your company. Managers should implement solutions that neutralize the retention risk caused by social media. They can create social media groups in which employees will be more likely to collaborate and less likely to share withdrawal intentions or discussions about external job opportunities. Managers can also use social media to directly reduce turnover (跳槽) intentions, by recognizing employees' accomplishments and giving visibility to employees' success stories.

1. What does previous research about social media reveal?

A. Most employees think positively of it.
B. It improves employees' work efficiency.
C. It enables employees to form connections.
D. Employees spend much of their work time on it.

2. What did the author's own research find about social media?

A. It influences employees' work negatively.
B. It does much harm to employee loyalty.
C. It kills employees' motivation for work.
D. It affects employers' decision-making.

3. What did the author find in his study about the effect of online social interactions?

A. It differs from employee to employee.
B. It tends to vary with the platform used.
C. It has much to do with whom employees interact with.
D. It is hard to measure when employees interact with outsiders.

4. What problem was found with employees using social media for work?

A. They seldom expressed their inner thoughts.
B. Most of them explored new job opportunities.
C. They were reluctant to collaborate with others.
D. Many of them ended with lower productivity.

5. What does the author suggest managers do to neutralize the retention risk?

A. Give promotions to employees for their accomplishments.
B. Create opportunities for employees to share success stories.
C. Acknowledge employees' achievements through social media.
D. Encourage employees to increase their visibility on social media.

Passage Two

Educators and business leaders have more in common than it may seem. Teachers want to prepare students for a successful future. Technology companies have an interest in developing a workforce with the STEM (science, technol-ogy, engineering and math) skills needed to grow the company and advance the industry. How can they wok together to achieve these goals? Play may be the answer.

Focusing on STEM skills is important, but the reality is that STEM skills are enhanced and more relevant when combined with traditional, hands-on creative activities. This combination is proving to be the best way to prepare today's children to be the makers and builders of tomorrow. That is why technology companies are partnering with educators to bring back good, old-fashioned play.

In fact many experts argue that the most important 2lst-century skills aren't related to specific technologies or subject matter, but to creativity; skills like imagination, problem-finding and problem-solving, teamwork, optimism, patience and the ability to experiment and take risks. These are skills acquired when kids tinker (鼓捣小玩意) , High-tech industries such as NASA's Jet Propulsion Laboratory have found that their best overall problem solvers were master tinkerers in their youth. There are cognitive (认知的) benefits of doing things the way we did as children —building something, tearing it down, then building it up again. Research shows that given 15 minutes of free play, four-and five-year-olds will spend a third of this time engaged in spatial, mathematical, and architectural activities. This type of play—especially with building blocks—helps children discover and develop key principles in math and geometry.

If play and building are critical to 21st century skill development, that's really good news for two reasons: Children are born builders, makers, and creators, so fostering（培养）21st century skills may be as simple as giving kids room to play, tinker and try things out, even as they grow older; Secondly, it doesn't take 21st century technology to foster 21st century skills. This is especially important for under-resourced schools and communities. Taking whatever materials are handy and tinkering with them is a simple way to engage those important "maker" skills. And anyone, anywhere, can do it.

1. What does the author say about educators?

A. They seek advice from technology companies to achieve teaching goals.
B. They have been successful in preparing the workforce for companies.
C. They help students acquire the skills needed for their future success.
D. They partner with technology companies to enhance teaching efficiency.

2. How can educators better develop students'STEM skills, according to the author?

A. By blending them with traditional, stimulating activities.
B. By inviting business leaders to help design curriculums.
C. By enhancing students'ability to think in a critical way.
D. By showing students the best way to learn is through play.

3. How do children acquire the skills needed for the 21 st century?

A. By engaging in activities involving specific technologies.
B. By playing with things to solve problems on their own.
C. By familiarizing themselves with high-tech gadgets.
D. By mastering basic principles through teamwork.

4. What can we do to help children learn the basics of math and geometry?

A. Stimulate their interest as early as possible.
B. Spend more time playing games with them.
C. Encourage them to make things with hands.
D. Allow them to tinker freely with calculators.

5. What does the author advise disadvantaged schools and communities to do?

A. Train students to be makers to meet future market demands.
B. Develop students'creative skills with the resources available.
C. Engage students with challenging tasks to foster their creativity.
D. Work together with companies to improve their teaching facilities.

Passage Three

In some countries, many universities use an employment system for teachers known as tenure. After a lengthy trial period, a faculty member whose performance meets with the approval of the senior members of the department and the administration of the institution may be awarded tenure. A tenured faculty member enjoys considerable job security for the rest of his or her working life and can only be fired for reasons of " moral turpitude" (bad or evil behavior) or " gross incompetence" or if the financial stability of the institution requires the elimination of an entire department or program.

The high degree of job security enjoyed by tenured faculty members has been the source of complaints about the tenure system. One issue that has been raised by many, including legislators evaluating the finances and managerial practices of state universities in the United States, is that tenure shelters faculty from accountability for poor performance. Another argument is that tenure makes the university inefficient in responding to changing instructional demands. It is difficult to substitute computer engineering faculty for civil engineering faculty if most of the latter have tenure. In 1988, the Education Reform Act significantly softened the tenure system in the United Kingdom, making it easier to fire individual faculty members for financial reasons. More recently, some universities in the United States have taken steps to give university administrators more control over tenured professors. And, in general, American institutions of higher learning have tended to increase the use of part-time and nontenured instructors over time. In 1992, just 48 percent of all instructors had tenure or were in a position that was expected to lead to tenure. The traditional argument in favor of tenure is based on academic freedom, the freedom to investigate and teach any area of knowledge without restriction or interference. In this view, tenure protects faculty members from retaliation for voicing unpopular views. For example, a labor economist might not present a complete examination of the costs and benefits of worker unions if he or she feared that a rabidly anti-union university leader might seek to have the economist fired for speaking of the positive aspects of unions. In fact, the American Association of University Professors (AAUP), a group dedicated to protecting academic freedom, got its start in the wake of a 1901 decision by Stanford University to fire economics instructor Edward Ross at the insistence of the university's co-founder, Jane Stanford, who objected to his views on economics and other matters.

Going beyond academic freedom, the economics literature has recently turned to an emphasis on tenure as a labor-market institution that may have a positive payoff to universities through the incentives (motivation) it provides.

1. Paragraph 1 answers which of the following questions about tenure?

A. How long must a teacher work before being considered for tenure?
B. What performance standards do senior faculty and administrators use to decide whether to award tenure?
C. In what situations may a tenured teacher lose his or her job?
D. How frequently are tenured teachers fired?

2. It can be inferred from paragraphs 1 and 2 that some critics of the tenure system believe that tenure can

A. reduce faculty members' motivation to do their jobs well
B. cause faculty members to prefer employment in state universities over employment in other universities
C. remove moral turpitude as a criterion for evaluating a faculty member's performance
D. encourage faculty members to switch from one department or program to another within a university

3. In paragraph 2, why does the author state that it may be difficult for some universities to substitute computer engineering faculty for tenured civil-engineering faculty?

A. To help explain why teachers in some fields lack as much job security as teachers in other fields do
B. To suggest a reason for the shortage of qualified professors in some fields but not in others
C. To identify a possible reason for poor performance by some academic departments
D. To help make clear the argument that tenure can decrease universities' ability to respond to changes in teaching needs

4. According to paragraph 2, all of the following steps have been taken in response to complaints about tenure EXCEPT?

A. passing a law that allows universities to fire professors in order to save money
B. providing administrators with more control of professors who have tenure
C. allowing each department to have the same percentage of tenured faculty
D. hiring more part-time teachers

5. Which of the following can be inferred from paragraph 3 about the 1901 decision by Stanford University to fire economics teacher Edward Ross?

A. It led to greater awareness of possible conflicts of interest between university administrators and university founders.

B. It stimulated efforts to protect the jobs of professors against threats to their academic freedom.

C. It revealed how interested American university administrators were in the views of their economics teachers.

D. It encouraged many American universities to begin hiring instructors who opposed the AAUP.

Passage Four

Journal editors decide what gets published and what doesn't, affecting the careers of other academics and influencing the direction that a field takes. You'd hope, then, that journals would do everything they can to establish a diverse editorial board, reflecting a variety of voices, experiences, and identities.

Unfortunately, a new study in *Nature Neuroscience* makes for disheartening reading. The team finds that the majority of editors in top psychology and neuroscience journals are male and based in the United States: a situation that may be amplifying existing gender inequalities in the field and influencing the kind of research that gets published.

Men were found to account for 60% of the editors of psychology journals. There were significantly more male than female editors at each level of seniority, and men made up the majority of editors in over three-quarters of the journals. Crucially, the proportion of female editors was significantly lower than the overall proportion of women psychology researchers.

The differences were even starker in the neuroscience journals: 70% of editors were male, and men held the majority of editorial positions in 88% of journals. In this case, the proportion of female editors was not significantly lower than the proportion of female researchers working in neuroscience —a finding that reveals enduring gender disparities in the field more broadly. Based on their results, the team concludes that "the ideas, values, and decision-making biases of men are over-represented in the editorial positions of the most recognized academic journals in psychology and neuroscience."

Gender inequality in science is often attributed to the fact that senior academics are more likely to be male, because historically science was male-dominated: it's argued that as time goes on and more women rise to senior roles, the field will become more equal. Yet this study showed that even the junior roles in psychology journals tended to be held disproportionately by men, despite the fact that there are actually more female than male junior psychology faculty. This implies that a lack of female academics is not the problem. Instead, there are structural reasons that women are disadvantaged in science. Women receive lower salaries and face greater childcare demands, for instance, which can result in fewer publications and grants—the kinds of things that journals look for when deciding who to appoint.

Rather than simply blaming the inequality of editorial boards on tradition, we should be actively breaking down these existing barriers. A lack of diversity among journal editors also likely contributes to psychology's WEIRD problem. If journal editors are largely men from the United States, then they will probably place higher value on papers that are relevant to Western, male populations, whether consciously or not.

1. What would we expect an editorial board of an academic journal to exhibit in view of its important responsibilities?

A. Insight. B. Expertise.

C. Integrity. D. Diversity.

2. What do we learn from the findings of a new study in *Nature Neuroscience*?

A. The majority of top psychology and neuroscience journals reflect a variety of voices, experiences, and identities.

B. The editorial boards of most psychology and neuroscience journals do influence the direction their field takes.

C. The editorial boards of the most important journals in psychology and neuroscience are male-dominated.

D. The majority of editors in top psychology and neuroscience journals have relevant backgrounds.

3. What fact does the author highlight concerning the gender differences in editors of psychology journals?

A. There were quite a few female editors who also distinguished themselves as influential psychology researchers.

B. The number of female editors was simply disproportionate to that of women engaged in psychology research.

C. The proportion of female editors was increasingly lower at senior levels.

D. There were few female editors who could move up to senior positions.

4. What can we infer from the conclusion drawn by the team of the new study on the basis of their findings?

A. Women's views are underrepresented in the editorial boards of top psychology and neuroscience journals.

B. Male editors of top psychology and neuroscience journals tend to be biased against their female colleagues.

C. Male researchers have enough representation in the editorial boards to ensure their publications.

D. Female editors have to struggle to get women's research articles published in academic journals.

5. What does the author suggest we do instead of simply blaming the inequality of editorial boards on tradition?

A. Strike a balance between male and female editors.

B. Increase women's employment in senior positions.

C. Enlarge the body of female academics.

D. Implement overall structural reforms.

Part III. Cloze Test（完形填空）

As a physician who travels quite a lot, I spend a lot of time on planes listening to that dreaded "Is there a doctor on board?" announcement. I've been ____71____ only once—for a woman who had merely fainted. But the ____72____ made me quite curious about how ____73____ this kind of thing happens. I wondered what I would do if ____74____ with a real mid-air medical emergency—without access ____75____ a hospital staff and the usual emergency equipment. So ____76____ the *New England Journal of Medicine* last week ____77____ a study about in-flight medical events, I read it ____78____ interest.

The study estimated that there are a(n) ____79____ of 30 in-flight medical emergencies on U.S. flights every day. Most of them are not ____80____; fainting and dizziness are the most frequent complaints. ____81____ 13% of them—roughly four a day—are serious enough to ____82____ a pilot to change course. The most common of the serious emergencies ____83____ heart trouble, strokes, and difficult breathing.

Let's face it: plane rides are ____84____. For starters, cabin pressures at high altitudes are set at roughly ____85____ they would be if you lived at 5,000 to 8,000 feet above sea level. Most people can tolerate these pressures pretty ____86____, but passengers with heart disease ____87____ experience chest pains as a result of the reduced amount of oxygen flowing through their blood. ____88____ common in-flight problem is deep venous thrombosis—the so-called economy class syndrome. ____89____ happens, don't panic. Things are getting better on the in-flight-emergency front. Thanks to more recent legislation, flights with at ____90____ one attendant are starting to install emergency medical kits to treat heart attacks.

71. A. called　　B. addressed
C. informed　　D. surveyed

72. A. accident B. condition
C. incident D. disaster

73. A. soon B. long
C. many D. often

74. A. confronted B. treated
C. identified D. provided

75. A. for B. to
C. by D. through

76. A. before B. since
C. when D. while

77. A. collected B. conducted
C. discovered D. published

78. A. by B. of
C. with D. in

79. A. amount B. average
C. sum D. number

80. A. significant B. heavy
C. common D. serious

81. A. For B. On
C. But D. So

82. A. require B. inspire
C. engage D. command

83. A. include B. confine
C. imply D. contain

84. A. enjoyable B. stimulating
C. tedious D. stressful

85. A. who B. what
C. which D. that

86. A. harshly B. reluctantly
C. easily D. casually

87. A. ought to B. may
C. used to D. need

88. A. Any B. One
C. Other D. Another

89. A. Whatever B. Whichever
C. Whenever D. Wherever

90. A. most B. worst
C. least D. best